

Order, order, order!

Ferromagnetism and the Curie temperature were explained by Weiss in terms of a huge internal ‘molecular field’ proportional to the magnetization. The theory is applicable both to localized and delocalized electrons. No such magnetic field really exists, but it is a useful way of approximating the effect of the interatomic Coulomb interaction in quantum mechanics, which Heisenberg described by the Hamiltonian $\mathcal{H} = -\mathcal{J}\mathbf{S}_1 \cdot \mathbf{S}_2$, where $\mathbf{S}_1$ and $\mathbf{S}_2$ are operators describing the localized spins on two adjacent atoms. When $\mathcal{J} > 0$, ferromagnetic exchange leads to ferromagnetic order in three dimensions. Spin waves are the low-energy excitations of the exchange-coupled magnetic lattice. In the delocalized electron picture, a ferromagnet has spontaneously spin-split energy bands. The density of $\uparrow$ and $\downarrow$ states is calculated using spin-dependent density functional theory. Important physical phenomena associated with ferromagnetism are discussed in this chapter, including magnetic anisotropy and, magnetoelastic, magneto-optic and magneto-transport effects.

The characteristic feature of a ferromagnet is its spontaneous magnetization M_s , which is due to alignment of the magnetic moments located on an atomic lattice. The magnetization tends to lie along easy directions determined by crystal structure, atomic-scale texture or sample shape. Heating above a critical temperature known as the Curie point, which ranges from less than 1 K for magnetically dilute salts to almost 1400 K for cobalt, leads to a reversible collapse of the spontaneous magnetization. Although there is no reason in principle why uniform ferromagnetic liquids should not exist, it seems that there are none. Ferrofluids, while ferromagnetic and liquid, are actually colloidal suspensions of solid ferromagnetic particles.

Important modifications of the electronic, thermal, elastic and optical properties are associated with magnetic order, whether ferromagnetic, or one of the more complex multisublattice or noncollinear ordered magnetic structures presented in the next chapter.

5.1 Mean field theory

5.1.1 Molecular field theory

The first modern theory of ferromagnetism, and one that remains useful today, was proposed by Pierre Weiss in 1906. Weiss's original theory was based on the classical paramagnetism of Langevin, but it was soon extended to the more general Brillouin theory of localized magnetic moments. His idea was that there is an internal 'molecular field' which is proportional to the magnetization of the ferromagnet. If n_W is the constant of proportionality, this adds to the internal contribution of any externally applied field:

$$H^i = n_W M + H. \quad (5.1)$$

H^i has to be enormous to induce a spontaneous magnetization at room temperature; the Weiss coefficient n_W is approximately 100. The magnetization is given by the Brillouin function (4.17) with $M_0 = nm_0 = ng\mu_B J$, where n is the number of magnetic atoms per unit volume,

$$M = M_0 \mathcal{B}_J(x), \quad (5.2)$$

but now

$$x = \mu_0 m_0 (n_W M + H) / k_B T. \quad (5.3)$$

In zero external field, M is the spontaneous magnetization M_s so we have

$$M_s / M_0 = \mathcal{B}_J(x_0), \quad (5.4)$$

where $x_0 = \mu_0 m_0 n_W M_s / k_B T$. Combining x_0 with $M_0 = nm_0$, we find $M_s / M_0 = (nk_B T / \mu_0 M_0^2 n_W) x_0$, which is conveniently written in terms of the Curie constant C (4.16) as

$$M_s / M_0 = [T(J + 1) / 3J C n_W] x_0. \quad (5.5)$$

The simultaneous solution of (5.4) and (5.5) is found graphically as indicated in Fig. 5.1. Otherwise the equations can be solved numerically. Results for M_s / M_0 versus T / T_C are plotted in Fig. 5.2 for some values of J , including the classical limit $J \rightarrow \infty$ where (5.4) is replaced by the Langevin function (4.21). In the Brillouin theory, the magnetization approaches zero temperature with horizontal slope, as required by thermodynamics (§2.5.4). Numerical values of the reduced spontaneous magnetization M_s / M_0 versus reduced temperature T / T_C are listed in Appendix G for different values of J . When S is the good quantum number, J in these formulae is replaced by S . Theory and experiment for nickel are compared in Fig. 5.3.

Weiss's molecular field theory was the first mean field theory of a phase transition. The moments are completely disordered at and above T_C , where the

Figure 5.1

Graphical solution of (5.4) and (5.5) for $J = \frac{1}{2}$ to find the spontaneous magnetization M_s when $T < T_C$. Equation (5.5) is also plotted for $T = T_C$ and $T > T_C$. The effect of an external field is to offset (5.5), as shown by the dotted line.

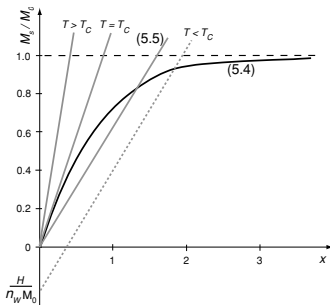
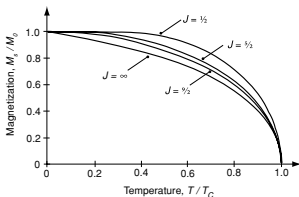


Figure 5.2

Spontaneous magnetization as a function of temperature calculated from molecular field theory, based on the Brillouin function for different values of J . The classical limit $J = \infty$ is based on the Langevin function.



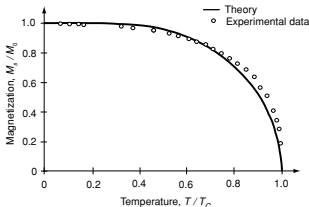
$2J + 1$ energetically degenerate M_J levels are equally populated. The magnetic entropy (4.25) then is $R \ln(2J + 1)$ per mole, where $R = N_A k_B$ is the gas constant, 8.315 J mol^{-1} . Below T_C , and especially just below, there is a specific heat of magnetic origin, as energy is absorbed to disorder the moments when the system is heated. A discontinuity in specific heat appears at T_C .

On a plot of M_s/M_0 versus x , the slope of (5.5) precisely at the Curie temperature is equal to the slope at the origin of the Brillouin function. For small x (4.19) $B_J(x) \approx [(J + 1)/3J]x$, hence there is a direct relation between Curie constant and Curie temperature:

$$T_C = n_W C. \quad (5.6)$$

Figure 5.3

The spontaneous magnetization for nickel, together with the theoretical curve for $J = \frac{1}{2}$ from the molecular field theory. Note that the theoretical curve is scaled to give correct values at each end.



In practice, T_C is used to determine n_W . Taking gadolinium as an example: $T_C = 292$ K, $J = S = 7/2$; $g = 2$; $n = 3.0 \times 10^{28} \text{ m}^{-3}$. Hence

$$C = \mu_0 n g^2 \mu_B^2 J(J+1)/3k_B \quad (4.16)$$

is 4.9 K, and the Weiss coefficient works out as $n_W = 59$.

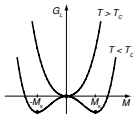
The paramagnetic susceptibility above T_C is obtained from (4.19), (5.3) and (5.4) in the small- x limit. The result is the Curie-Weiss law

$$\chi = C/(T - \theta_p), \quad (5.7)$$

where

$$\theta_p = T_C = \mu_0 n_W n g^2 \mu_B^2 J(J+1)/3k_B. \quad (5.8)$$

The Curie constant C is often written in terms of the effective moment m_{eff} as $C = \mu_0 n m_{\text{eff}}^2 / 3k_B$, where $m_{\text{eff}} = g \sqrt{J(J+1)} \mu_B$. In this theory, the paramagnetic Curie temperature θ_p is equal to the Curie temperature T_C , which is the point where the susceptibility diverges.



The Landau free energy for a ferromagnet at temperatures close to the Curie temperature. There are two energy minima at $\pm M_s$ for $T < T_C$, but a single minimum at $M = 0$ for $T > T_C$.

5.1.2 Landau theory

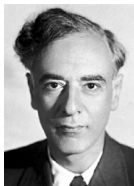
An approach that is equivalent to molecular field theory close to T_C , where M is small and aligned with any field external H' , is to expand the free energy G_L in even powers of M . Only even powers are permitted in the series, because time reversal symmetry requires that the energy is unchanged on reversing the magnetization, $G_L(M) = G_L(-M)$ in the absence of the external field:

$$G_L = AM^2 + BM^4 + \dots - \mu_0 H' M. \quad (5.9)$$

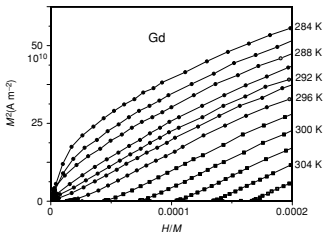
The coefficients A and B depend on temperature. There is a difference between the Landau free energy $G_L = f(M, T) - \mu_0 H' M$ and the Gibbs free energy $G(H', T) = F(M, T) - \mu_0 H' M$ (§2.5.4), where M is expressed as a function

Figure 5.4

Arrott Below plots to determine T_C for gadolinium. The experimentally measured magnetization is $\sigma = M d$, where d is the density, rather than M . (Data courtesy of M. Venkatesan.)



Lev Landau 1908–1968.



of the variables H', T via the equation of state $M = M(H', T)$; G_L is the energy of the state when M is forced to adopt a particular value, as if it were an external constraint. G_L is minimized in a local energy minimum with that value of M , which makes the approach useful for treating problems of hysteresis.

For $T < T_C$, energy minima at $M = \pm M_s$ imply $A < 0$ and $B > 0$. For $T > T_C$ an energy minimum at $M = 0$ implies $A > 0$ and $B > 0$. It follows that A must change sign at T_C . It has the form $a(T - T_C)$, where a is a constant independent of temperature, $a > 0$. The equilibrium magnetization minimizes G_L with respect to M ; $\partial G_L / \partial M = 0$ implies

$$2AM + 4BM^3 = \mu_0 H'. \quad (5.10)$$

Close to T_C , in zero field, $M_s^2 = -A/2B$, hence

$$M_s \approx \sqrt{a/2B}(T_C - T)^{1/2}, \quad (5.11)$$

as shown in Fig. 5.2. Ignoring the demagnetizing field, the Curie–Weiss susceptibility M/H' is given by (5.10) as $\mu_0/2A$;

$$\chi \approx (\mu_0/2a)(T - T_C)^{-1}. \quad (5.12)$$

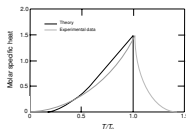
When the system is at a temperature exactly equal to T_C , $A = 0$ and (5.10) gives the critical isotherm

$$M = (\mu_0/4B)^{1/3} H'^{1/3}, \quad (5.13)$$

whereas in the vicinity of T_C (5.10) gives

$$M^2 = (\mu_0/4B)H'/M - (a/2B)(T - T_C). \quad (5.14)$$

This last equation is the basis of Arrott–Below plots used for precise determination of the Curie temperature. The $M(H)$ curves at different temperatures are plotted



Comparison of the measured magnetic specific heat of a ferromagnet near T_c (dotted line) with the prediction of mean field theory (solid line).

Table 5.1. The critical exponents of a mean field ferromagnet

Specific heat	$C_m \sim T_c - T ^\alpha$	$\alpha = 0$
Magnetization	$M_s \sim (T_c - T)^\beta$	$\beta = \frac{1}{2}$
Susceptibility	$\chi \sim (T - T_c)^{-\gamma}$	$\gamma = 1$
Critical isotherm	$M_s \sim H^{1/\delta}$	$\delta = 3$

as M^2 versus H'/M , and the isotherm that extrapolates to zero is the one at T_c (Fig. 5.4).

The magnetic specific heat C_m can also be calculated from Landau theory using $C_m = -T(\partial^2 G_L / \partial T^2)$. Results from (5.9) and (5.14) are $C_m = Ta^2/2B$ when $T = T_c^-$ and $C_m = 0$ when $T = T_c^+$. There is a stepwise discontinuity at T_c , with no magnetic specific heat above the transition where $M = 0$.

The Landau theory can be adapted to any continuous or discontinuous phase transition. M is the order parameter for the ferromagnet, H' is the conjugate field and the relation between them is the generalized susceptibility χ . Whatever the interpretation of these parameters in different physical systems, the power laws describing their variations with T near T_c are exactly the same. The same powers are obtained from Landau theory and from Weiss's molecular field theory. This can be verified by expanding the Brillouin function to order x^3 (4.19), which gives an expression equivalent to (5.9). Both are mean field theories of ferromagnetism. Other terms may be added to the free energy to include additional fields such as pressure or stress. It is remarkable how many relations can be established between different measurable physical quantities from an expansion of the free energy in powers of the order parameter.

The power law variations of the physical properties in the vicinity of T_c are summarized in Table 5.1. The values of the static critical exponents $\alpha, \beta, \gamma, \delta$ are common to all mean field theories.

Experimentally, the properties of ferromagnets do show power law behaviour in $(T - T_c)$ provided measurements are made sufficiently close to the Curie point, but the critical exponents are somewhat different from those predicted by mean field theory. For example, ferromagnets usually show a λ -type anomaly in their specific heat at T_c , rather than a stepwise discontinuity. The divergence is described by a critical exponent $\alpha \approx 0.1$, rather than zero. The residual magnetic specific heat above T_c is witness to the persistence of short-range order, which is not predicted by the theory. Above T_c , the susceptibility follows a power law $\chi \sim (T - T_c)^{-\gamma}$, where γ is about 1.3, whereas in mean field theory γ is 1 (the Curie–Weiss law). The critical exponents $\alpha, \beta, \gamma, \delta$ for nickel, for example, are 0.10, 0.42, 1.32 and 4.5, respectively. We return to this topic at the end of Chapter 6.

The other place where significant deviations from the mean field theory are found is at low temperatures, where the spin-wave excitations discussed later in the chapter are important.

Table 5.2. Dimensionless susceptibility of some metals at 298 K (units: 10^{-6})

Li	14	Sc	263	Cu	-10	Ce	1778
K	6	Y	121	Zn	-16	Nd	3433
Be	-24	Ti	182	Au	-34	Eu	15570
Ca	22	Nb	237	Al	21	Gd	476300
Ba	7	Mo	123	Sn	-29	Dy	68400
		Pd	805	Bi	-164	Tm	17710
		Pt	279				

5.1.3 Stoner criterion

The starting point for a discussion of ferromagnetism in metals is the band paramagnetism introduced in §3.2.6. The Pauli susceptibility is a small, positive quantity, practically independent of temperature because delocalized electrons obey Fermi–Dirac statistics; only the small fraction of them with energy close to ε_F are able to respond to a change in temperature or magnetic field.

In the three-dimensional free-electron model, the density of states $\mathcal{D}(\varepsilon)$ (states $\text{m}^{-3} \text{J}^{-1}$) varies as $\sqrt{\varepsilon}$ (3.39), and the $\uparrow$ and $\downarrow$ bands shift by $\mp \mu_0 H \mu_B$ in the field as shown in Fig. 3.7. The resulting susceptibility (3.43) can be written

$$\chi_P = \mu_0 \mu_B^2 \mathcal{D}(\varepsilon_F), \quad (5.15)$$

where $\mathcal{D}(\varepsilon_F)$ is the density of states at the Fermi level for both spins, which is double the density of states for one spin $\mathcal{D}_{\uparrow, \downarrow}(\varepsilon_F)$. The Pauli susceptibility is approximately 10^{-5} for many metals, but it approaches 10^{-3} for the 4d metal Pd (Table 5.2). Narrower bands tend to have higher susceptibility, because the density of states at ε_F scales as the inverse of the bandwidth. When the density of states is high enough, it becomes energetically favourable for the bands to split, and the metal becomes spontaneously ferromagnetic.

Stoner applied Weiss's molecular field idea to the free-electron gas. Assuming the linear variation of internal field with magnetization has a coefficient n_S :

$$H^i = n_S M + H, \quad (5.16)$$

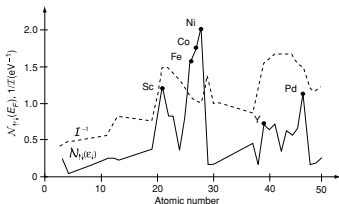
the Pauli susceptibility in the internal field is $\chi_P = M/(n_S M + H)$. Hence, the response to the field H

$$\chi = M/H = \chi_P/(1 - n_S \chi_P) \quad (5.17)$$

is a susceptibility that is enhanced when $n_S \chi_P < 1$ and diverges when $n_S \chi_P = 1$. Stoner expressed this condition in terms of the local density of states at the Fermi level, $\mathcal{D}(\varepsilon_F)$. Writing the exchange energy (in J m^{-3}) $-\frac{1}{2} \mu_0 H^i M = -\frac{1}{2} \mu_0 n_S M^2$ as $-(I/4)(n^\uparrow - n^\downarrow)^2/n$, where $M = (n^\uparrow - n^\downarrow)\mu_B$

Figure 5.5

Comparison of $\mathcal{N}_{\uparrow,\downarrow}(\varepsilon_F)$ with $1/\mathcal{I}$ for metallic elements.



and n is the number of atoms per unit volume, it follows from (5.15) that $n_S \chi_P = \mathcal{I} D(\varepsilon_F)/2n$. The metal becomes spontaneously ferromagnetic when the susceptibility diverges spontaneously; in other words when

$$\mathcal{I} N_{\uparrow,\downarrow}(\varepsilon_F) > 1, \quad (5.18)$$

where $\mathcal{N}_{\uparrow,\downarrow}(\varepsilon) = D(\varepsilon)/2n$ is the density of states per atom for each spin state. This is the famous Stoner criterion. The Stoner exchange parameter $\mathcal{I}$ is roughly 1 eV for the 3d ferromagnets, and $n_S \gtrsim 10^3$ for spontaneous band splitting. The exchange parameter has to be comparable to the width of the band for spontaneous splitting to be observed. Ferromagnetic metals have narrow bands and a peak in the density of states $N(\varepsilon)$ at or near ε_F . The data in Fig. 5.5 show that only Fe, Co and Ni meet the Stoner criterion. Pd comes close.

5.2 Exchange interactions

The origin of the effective field $\mathbf{H}^i$ is the exchange interaction, which reflects the Coulomb repulsion of two nearby electrons, usually on neighbouring atoms, acting in conjunction with the Pauli principle, which forbids the two electrons to enter the same quantum state. Electrons cannot be in the same place if they have the same spin. There is an energy difference between the $\uparrow_i \uparrow_j$ and $\uparrow_i \downarrow_j$ configurations of the spins of neighbouring atoms i, j . Interatomic exchange in insulators is usually one or two orders of magnitude weaker than the ferromagnetic intra-atomic exchange between electrons on the same atom, which leads to Hund's first rule.

As stated in §4.1, the Pauli principle forbids more than one electron to enter a quantum state, denoted by a particular set of quantum numbers. Electrons are indistinguishable, so exchange of two electrons must give the same electron density $|\Psi(1, 2)|^2 = |\Psi(2, 1)|^2$. Since electrons are fermions, the only solution

is for the total wave function of the two electrons to be antisymmetric

$$\Psi(1, 2) = -\Psi(2, 1). \quad (5.19)$$

The total wave function Ψ is the product of functions of space and spin coordinates $\phi(\mathbf{r}_1, \mathbf{r}_2)$ and $\chi(s_1, s_2)$.

The simple example of the hydrogen molecule H_2 with two atoms each having an electron in a hydrogenic $1s$ -orbital $\psi_i(\mathbf{r}_i)$ gives an idea of the physics of exchange. Schrödinger's equation is $\mathcal{H}(\mathbf{r}_1, \mathbf{r}_2)\Psi(\mathbf{r}_1, \mathbf{r}_2) = \varepsilon\Psi(\mathbf{r}_1, \mathbf{r}_2)$ where, neglecting the interactions between the electrons,

$$\left[-\frac{\hbar^2}{2m} \left(\frac{\partial^2}{\partial r_1^2} + \frac{\partial^2}{\partial r_2^2} \right) - \frac{e^2}{4\pi\epsilon_0} \left(\frac{1}{r_1} + \frac{1}{r_2} \right) \right] \Psi(\mathbf{r}_1, \mathbf{r}_2) = \varepsilon\Psi(\mathbf{r}_1, \mathbf{r}_2). \quad (5.20)$$

There are two molecular orbits, a spatially symmetric bonding orbital ϕ_s , with electronic charge piled up between the atoms, and a spatially antisymmetric antibonding orbital ϕ_a having a nodal plane with no charge midway between them. Chemical bonds which involve hybridized wave functions of electrons of neighbouring atoms are generally classified in this way:

$$\phi_s = (1/\sqrt{2})(\psi_1 + \psi_2) \quad \phi_a = (1/\sqrt{2})(\psi_1 - \psi_2). \quad (5.21)$$

ψ_1 and ψ_2 are the spatial components of the individual wave functions of electrons 1 and 2 respectively. The wave functions $\psi_1(\mathbf{r}_1)$ and $\psi_2(\mathbf{r}_2)$ are the solutions of Schrödinger's equation for each individual atom.

The symmetric and antisymmetric spin functions are the spin triplet and spin singlet states:

$$S = 1; \quad M_S = 1, 0, -1$$

$$\chi_s = |\uparrow_1, \uparrow_2\rangle; \quad (1/\sqrt{2})[|\uparrow_1, \downarrow_2\rangle + |\downarrow_1, \uparrow_2\rangle]; \quad |\downarrow_1, \downarrow_2\rangle$$

$$S = 0; \quad M_S = 0$$

$$\chi_a = (1/\sqrt{2})[|\uparrow_1, \downarrow_2\rangle - |\downarrow_1, \uparrow_2\rangle]$$

According to (5.19), the symmetric space function must multiply the antisymmetric spin function, and vice versa. Hence the total antisymmetric wave functions are

$$\Psi_I = \phi_s(1, 2)\chi_a(1, 2),$$

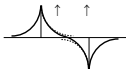
$$\Psi_{II} = \phi_a(1, 2)\chi_s(1, 2).$$

When the two electrons are in a spin triplet state, there can be no chance of finding them at the same point of space. Electrons with parallel spins avoid each other. But if the electrons are in the spin singlet state, with antiparallel spins, there is some probability of finding them in the same place, because the spatial part of the wave function is symmetric under exchange of the electrons.

$S = 0$



$S = 1$



The spatially symmetric and antisymmetric wave functions for the H_2 molecule.

The energies of the two states can be evaluated from the Hamiltonian $\mathcal{H}(\mathbf{r}_1, \mathbf{r}_2)$ in (5.20):

$$\varepsilon_{I,II} = \int \phi_{s,a}^*(\mathbf{r}_1, \mathbf{r}_2) \mathcal{H}(\mathbf{r}_1, \mathbf{r}_2) \phi_{s,a}(\mathbf{r}_1, \mathbf{r}_2) d\mathbf{r}_1^3 d\mathbf{r}_2^3.$$

For the hydrogen molecule, ε_I is lower than ε_{II} . In other words, the bonding orbital/spin singlet state lies below the antibonding orbital/spin triplet state because of the spatial constraint on the triplet. Setting the exchange integral $\mathcal{J} = (\varepsilon_I - \varepsilon_{II})/2$, we can write the energy in the form

$$\varepsilon = -2(\mathcal{J}/\hbar^2) \mathbf{s}_1 \cdot \mathbf{s}_2, \quad (5.22)$$

where the product $\mathbf{s}_1 \cdot \mathbf{s}_2$ is $\frac{1}{2}[(\mathbf{s}_1 + \mathbf{s}_2)^2 - s_1^2 - s_2^2]$. According to whether the spin quantum number $S = s_1 + s_2$ is 0 or 1, the eigenvalues are $-\frac{3}{4}\hbar^2$ or $+\frac{1}{4}\hbar^2$. The energy splitting between the singlet state Ψ_I and the triplet state Ψ_{II} is $2\mathcal{J}$. Here $\mathcal{J}$ is the exchange integral

$$\mathcal{J} = \int \psi_1^*(\mathbf{r}') \psi_2^*(\mathbf{r}) \mathcal{H}(\mathbf{r}, \mathbf{r}') \psi_1(\mathbf{r}) \psi_2(\mathbf{r}') d\mathbf{r}^3 d\mathbf{r}'^3.$$

In the H_2 molecule, the spin singlet state is lower, so the integral is negative. In an atom, however, the orbitals are orthogonal and $\mathcal{J}$ is positive.

Heisenberg generalized (5.22) to many-electron atomic spins $\mathbf{S}_1$ and $\mathbf{S}_2$, writing his famous Hamiltonian

$$\mathcal{H} = -2\mathcal{J} \hat{\mathbf{S}}_1 \cdot \hat{\mathbf{S}}_2, \quad (5.23)$$

where $\hat{\mathbf{S}}_1$ and $\hat{\mathbf{S}}_2$ are dimensionless spin operators, like the Pauli spin matrices in (3.17). The $\hbar^2$ has been absorbed into the exchange constant $\mathcal{J}$, which has units of energy. From now on we will adopt this convention, in order to avoid writing $\hbar$ everywhere. We also drop the hat on the spin operators, $\hat{\mathbf{S}}_i$. The exchange integral $\mathcal{J}$ then has dimensions of energy, and it is often expressed in kelvins by dividing it by k_B , Boltzmann's constant. $\mathcal{J} > 0$ indicates a ferromagnetic interaction, which tends to align the two spins parallel; $\mathcal{J} < 0$ indicates an antiferromagnetic interaction, which tends to align the two spins antiparallel.

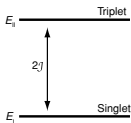
When there is a lattice, the Hamiltonian¹ is generalized to a sum over all pairs of atoms on lattice sites i, j :

$$\mathcal{H} = -2 \sum_{i>j} \mathcal{J}_{ij} \mathbf{S}_i \cdot \mathbf{S}_j. \quad (5.24)$$

This is simplified to a sum with a single exchange constant $\mathcal{J}$ if only nearest-neighbour interactions count. The interatomic exchange coupling described by the Heisenberg Hamiltonian can only be ferromagnetic or antiferromagnetic.

The Heisenberg exchange constant $\mathcal{J}$ can be related to the Weiss constant n_W of the molecular field theory. Suppose that a moment $g\mu_B \mathbf{S}_i$ interacts with an effective field $H^i = n_W M = n_W n g \mu_B S$, and that in the Heisenberg model

¹ Other conventions exist, omitting the 2 and/or counting each pair in the sum twice.



Splitting of the spin singlet and spin triplet states for the H_2 molecule. The exchange integral $\mathcal{J}$ is negative, so the singlet is lower.



Junjiro Kanamori, 1930–.

only the nearest neighbours of S_i interact appreciably with it. Then the site Hamiltonian is

$$\mathcal{H}_i = -2 \left[\sum_j \mathcal{J} S_j \right] \cdot S_i \approx -\mu_0 H^i g \mu_B S_i. \quad (5.25)$$

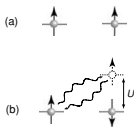
The molecular field approximation amounts to averaging out the local correlations between S_i and S_j . If Z is the number of nearest neighbours in the sum, then $\mathcal{J} = \mu_0 n w n g^2 \mu_B^2 / 2Z$. Hence, from (5.8)

$$T_C = \frac{2Z\mathcal{J}S(S+1)}{3k_B}. \quad (5.26)$$

Taking the example of gadolinium again, where $T_C = 292$ K, $S = 7/2$, $Z = 12$, we find $\mathcal{J}/k_B = 2.3$ K.

The Heisenberg Hamiltonian (5.23) indicates that exchange interactions couple the atomic *spins*. It can be applied directly to the 3d elements, where the crystal field ensures that spin is a good quantum number, and to the rare-earth ions Eu^{2+} and Gd^{3+} , which have no orbital moment. However, J is the good quantum number for the other rare-earths, so S must be projected onto J , as explained below. Ions with a $J = 0$ ground-state multiplet, Sm^{2+} and Eu^{3+} , cannot order magnetically despite their large spin quantum number, $S = 3$.

Generally, the energy of any electronic system is lowered as the wave functions spread out. This follows from the uncertainty principle $\Delta p \Delta x \approx \hbar$. When many more-or-less delocalized electrons are present in different orbitals, the calculation of exchange is a delicate matter. Orbital degeneracy, absent in the H_2 molecule, opens the possibility that triplet states may be lower in energy than singlets. The energies involved are only ≈ 1 meV, compared with bandwidths of order 1–10 eV. Competing exchange interactions may coexist with different signs of coupling. It is therefore best to describe exchange phenomenologically, and determine the exchange interactions experimentally.



The antiferromagnetic superexchange interaction. Two neighbouring sites with singly occupied orbitals are shown with (a) parallel or (b) antiparallel spin alignment. Hopping is forbidden by the Pauli principle in the parallel case. There is an energy gain due to virtual hopping in the antiparallel case.

5.2.1 Exchange in insulators

Superexchange The electrons in insulators are localized. Oxides are a good example. There is little direct 3d–3d overlap in transition-metal oxides, but the 3d-orbitals are hybridized with the oxygen 2p-orbitals; $\phi_{3d} = \alpha\psi_{3d} + \beta\psi_{2p}$ with $|\alpha|^2 + |\beta|^2 = 1$. The oxygen bridges transmit a ‘superexchange’ interaction, which can be described by the Heisenberg Hamiltonian.

Figure 5.6 shows a typical superexchange bond. In the case of a singly occupied 3d-orbital or a half-filled d shell (Fe^{3+} , Mn^{2+}), configuration (b) is lower in energy than configuration (a) because both electrons in an oxygen 2p-orbital can then spread out into unoccupied 3d-orbitals. The superexchange interaction $\mathcal{J}$ involves simultaneous virtual transfer of two electrons with the instantaneous formation of a $3d^{n+1}2p^5$ excited state; the interaction is of order $-2t^2/U$, where t is the p – d transfer integral and U is the on-site 3d Coulomb

Figure 5.6

A typical superexchange bond. Configuration (b) is lower in energy than configuration (a).



John B. Goodenough, 1922–.

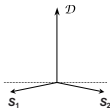


(a)

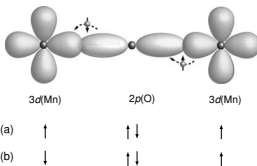


(b)

Overlapping d -orbitals characterized by (a) nonzero and (b) zero overlap integrals. Dark and light shading denotes positive and negative sign of the wave function.



Canted antiferromagnetism due to the Dzyaloshinski-Moriya interaction.



interaction. The transfer integral is of order 0.1 eV and the on-site Coulomb interaction is in the range 3–5 eV. $\mathcal{J}$ depends sensitively on the interatomic separation, but also on the M–O–M bond angle, varying as $\cos^2 \theta_{12}$.

The occupancy and orbital degeneracy of the $3d$ states is the critical factor in determining the strength and sign of superexchange. There are many possible cases to consider and the results were summarized in the Goodenough-Kanamori rules. The rules were reformulated by Anderson, in a simpler way that makes it unnecessary to consider the oxygen.

- (i) When two cations have lobes of singly occupied $3d$ -orbitals which point towards each other giving large overlap and hopping integrals, the exchange is strong and *antiferromagnetic* ($\mathcal{J} < 0$). This is the usual case, for 120 – 180° M–O–M bonds.
- (ii) When two cations have an overlap integral between singly occupied $3d$ -orbitals which is zero by symmetry, the exchange is *ferromagnetic* and relatively weak. This is the case for $\sim 90^\circ$ M–O–M bonds.
- (iii) When two cations have an overlap between singly occupied $3d$ orbitals and empty or doubly occupied orbitals of the same type, the exchange is also *ferromagnetic*, and relatively weak.

Superexchange is more commonly antiferromagnetic than ferromagnetic, because the overlap integrals are more likely to be large than zero.

Antisymmetric exchange A few materials with low symmetry exhibit a weak antisymmetric coupling, the Dzyaloshinski-Moriya interaction. This is represented by the Hamiltonian

$$\mathcal{H} = -\mathcal{D} \cdot (\mathbf{S}_i \times \mathbf{S}_j), \quad (5.27)$$

where $\mathcal{D}$ is a vector which lies along a high-symmetry axis, so the tendency is to couple the two spins *perpendicularly*. This is a higher-order effect, occurring between ions already coupled by superexchange; $|\mathcal{D}/\mathcal{J}| \approx 10^{-2}$. In an antiferromagnet, the spins may be canted away from the antiferromagnetic axis by about 1° . Antisymmetric exchange is the reason why antiferromagnets with a uniaxial crystal structure such as MnF_2 , MnCO_3 and $\alpha\text{Fe}_2\text{O}_3$ may exhibit a weak ferromagnetic moment. In the older literature the term parasitic

ferromagnetism is encountered for this kind of intrinsic weak ferromagnetism, because it was thought to be due to ferromagnetic impurities. A moment only appears when the antiferromagnetic axis is perpendicular to the crystallographic axis of symmetry, along which $\mathcal{D}$ is constrained to lie. It disappears when the axes are parallel.

Biquadratic exchange This is another weak, higher-order effect which is sometimes detectable for the rare-earths. It is represented by the Hamiltonian

$$\mathcal{H} = -\mathcal{B}(\mathbf{S}_i \cdot \mathbf{S}_j)^2. \quad (5.28)$$

5.2.2 Exchange in metals

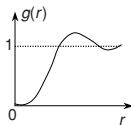
The principal exchange mechanism in ferromagnetic and antiferromagnetic metals involves overlap of the partly localized atomic orbitals of adjacent atoms. Other exchange mechanisms involve the interaction of purely delocalized electrons or of localized and delocalized electrons in the metal.

Direct exchange In 3d metals, the electrons are described by extended wave functions and a spin-polarized local density of states. It is usually more appropriate to describe them by the one-electron d wave functions of §4.4.2, rather than the free-electron waves of §3.2.5. In the tight-binding model the overlap of the one-electron wave functions is small and the electrons remain mostly localized on the atoms. The model Hamiltonian is

$$\mathcal{H} = \sum_{ij} t_{ij} c_i^\dagger c_j,$$

where the sum represents the conduction band in terms of the electron creation and annihilation operators² $c^\dagger$ and c . Usually only nearest-neighbour interactions are important and the interatomic transfer integral $t_{ij} = t$. The bandwidth in the tight-binding model is $W = 2Zt$, where Z is the number of nearest neighbours. In 3d metals $t \approx 0.1$ eV and $Z = 8 - 12$, so the d bands are a few eV wide. Exchange in a roughly half-filled band is antiferromagnetic, because the energy gain associated with letting the wave functions expand onto neighbouring sites is only achieved when the neighbours are antiparallel, leaving empty $\uparrow$ orbitals on the neighbouring sites to transfer into. Nearly filled or nearly empty bands tend to be ferromagnetic (Fig. 5.7) because electrons can then hop into empty states with the same spin. This helps to explain why chromium and manganese are antiferromagnetic, but iron, cobalt and nickel are ferromagnetic.

Bandwidth is the enemy of exchange. As t becomes large, the electrons are delocalized regardless of their spin. The alkali metals, for example, are Pauli

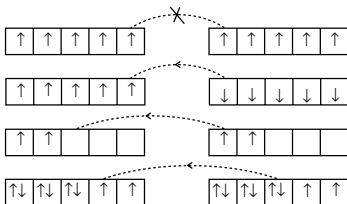


The exchange hole: the normalized probability of finding two electrons with the same spin a distance r apart.

² c_j is an operator that destroys an electron on site j , while $c_i^\dagger$ is an operator that creates an electron on site i . The product $c_i^\dagger c_j$ therefore transfers an electron from site j to site i .

Figure 5.7

Electron delocalization in d bands which are half-full, almost empty or almost full.



paramagnets described by a free-electron model with one electron per atom. The early $3d$ metals scandium, titanium and vanadium are not ferromagnetic because t is too big. Scandium comes close. If it were possible to dilate the lattice to reduce t a little, scandium would become ferromagnetic.

The sign of the direct exchange depends principally on band occupancy, and then on the interatomic spacing, with ferromagnetic exchange favoured at larger spacing. The exchange is greatest just after the critical condition for the appearance of magnetism, $U/W > (U/W)_{crit}$, where U is the on-site Coulomb interaction, and W is the bandwidth.

s - d model Coupling of the spins s of the conduction electrons with core spins S in a metal is generally represented by a Hamiltonian including the term

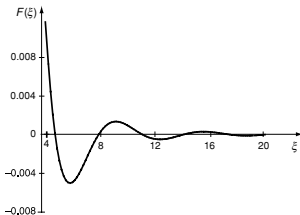
$$-\mathcal{J}_{sd}\Omega|\psi|^2\mathbf{S}\cdot\mathbf{s}, \quad (5.29)$$

where Ω is the volume of the core d shell and $|\psi|^2$ is the s -electron probability density. The s - d coupling is an on-site interaction, so the coupling constant is large, $\mathcal{J}_{sd} \approx 1$ eV. This interaction may lead to long-range ferromagnetic coupling between the core spins, regardless of whether $\mathcal{J}_{sd}$ is positive or negative. The host conduction band is supposed to be uniformly spin-polarized parallel or antiparallel to the core spins.

RKKY interaction The ' s - d ' model applies as well to rare-earths, where the core spins are not $3d$, but $4f$. The localized moments in the $4f$ shell interact via electrons in the $5d/6s$ conduction band. The on-site interaction between a core spin S and a conduction electron spin s is $-\mathcal{J}_{sf}\mathbf{S}\cdot\mathbf{s}$, where $\mathcal{J}_{sf} \approx 0.2$ eV. Ruderman, Kittel, Kasuya and Yosida showed that a single magnetic impurity actually creates a nonuniform, oscillating spin polarization in the conduction band which falls off as r^{-3} . This spin polarization is related to the Friedel oscillations of charge density around the impurity which have wavelength π/k_F . It leads to long-range oscillatory coupling between core spins. For free

Figure 5.8

The RKKY function $F(\xi)$. Note that $F(\xi)$ becomes very large when $\xi < 4$.



electrons, the polarization is proportional to the RKKY function

$$F(\xi) = (\sin \xi - \xi \cos \xi) / \xi^4,$$

where $\xi = 2k_F r$, k_F being the Fermi wavevector (Fig. 5.8). This oscillating spin polarization results from the different potential seen by the $\uparrow$ and $\downarrow$ conduction electrons at the local moment site. The first zero of $F(\xi)$ is at $\xi = 4.5$. The effective coupling between two localized spins is

$$\mathcal{J}_{\text{eff}} \approx \frac{9\pi \mathcal{J}_{sf}^2 v^2 F(\xi)}{64 \mathcal{E}_F}, \quad (5.30)$$

where v is the number of conduction electrons per atom and $\mathcal{E}_F$ is the Fermi energy. Since the Fermi wavevector is about 0.1 nm^{-1} (Table 3.3), the sign of $\mathcal{J}_{\text{eff}}$ fluctuates on a scale of nanometres. When only ferromagnetic nearest-neighbour coupling is important, the Curie temperature can be deduced from (5.26). The RKKY interaction in the low-electron-density limit is equivalent to the s - d model with ferromagnetic coupling. Analogous oscillatory exchange is found in ferromagnetic multilayers with nonmagnetic spacer layers.

Among the rare-earth metals, only gadolinium has S as a good quantum number. The others have J as their quantum number, yet the exchange interaction couples spins. We therefore need to project S onto J when calculating the exchange coupling, whether direct or indirect. Since $\mathbf{L} + 2\mathbf{S} = g\mathbf{J}$ and $\mathbf{J} = \mathbf{L} + \mathbf{S}$, $\mathbf{S} = (g - 1)\mathbf{J}$. This introduces a factor $(g - 1)^2 J(J + 1)$ into the exchange coupling. The factor is squared, because the spin enters the exchange interaction between two rare-earths twice. The effective coupling is

$$\mathcal{J}_{\text{RKKY}} = G \mathcal{J}_{\text{eff}}.$$

where $G = (g - 1)^2 J(J + 1)$ is the de Gennes factor. Magnetic ordering temperatures for any series of rare-earth metals or compounds with the same conduction-band structure and similar lattice spacings should scale with G ,

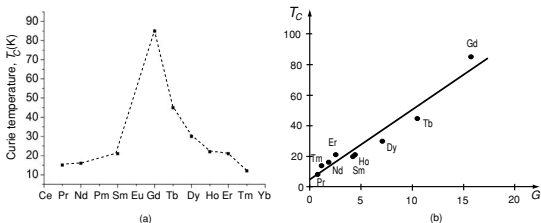


Figure 5.9

(a) Curie temperatures of ferromagnetic RNi_2 compounds; (b) a plot of T_C versus the de Gennes factor G .

and show a maximum for gadolinium. The de Gennes factor was included in Table 4.10. Figure 5.9 displays data for the Curie temperature of a series of ferromagnetic RNi_2 compounds. When plotted versus G , the data follow a straight line. The nickel in this series of intermetallic compounds is nonmagnetic.

Double exchange This interaction arises between $3d$ ions which have both localized and delocalized d electrons. Unlike ferromagnetic superexchange, mixed valence configurations are required for double exchange, as they are in any metal, but unlike a normal metal, the number of configurations is restricted to just two. In copper, for example, with its one electron in a broad $4s$ band, the instantaneous atomic configurations are s^0 , s^1 and s^2 . Electronic correlations are weak in broad bands, so the three configurations will appear with probabilities of $\frac{1}{4}$, $\frac{1}{2}$ and $\frac{1}{4}$. By contrast, a double-exchange material, such as the manganite $(\text{La}_{0.7}\text{Ca}_{0.3})\text{MnO}_3$, has both Mn^{4+} and Mn^{3+} ions (d^3 and d^4) present on octahedral sites. The two Mn valence states are imposed by the charge states of the other ions in the compound, La^{3+} , Ca^{2+} and O^{2-} . The d^3 core electrons for both octahedrally coordinated ions are localized in a narrow $t_{2g}^{\uparrow}$ band, but the fourth d electron inhabits a broader $e_g^{\uparrow}$ band, hybridized with oxygen, where it can hop from one d^3 core to another, Fig. 5.10. The configurations $d_i^3 d_j^4$ and $d_i^4 d_j^3$ on adjacent sites i and j are practically degenerate. On each site, there is strong on-site Hund's rule exchange coupling $\mathcal{J}_H \approx 2$ eV between t_{2g} and e_g electrons. Electrons can hop freely if the core spins are parallel, but when they are antiparallel there is a large energy barrier due to the Hund's rule interaction. If the quantization axes of adjacent sites are misaligned by an angle θ , the eigenvector of a $\uparrow$ electron in the rotated frame is $\begin{pmatrix} \cos \theta/2 \\ \sin \theta/2 \end{pmatrix}$ (3.24). The transfer integral t therefore varies as $\cos(\theta/2)$. Double exchange is ferromagnetic because the transfer is zero when the ions on adjacent sites are antiparallel, $\theta = \pi$.

Figure 5.10

The double-exchange interaction. The electron hops with spin memory from one localized ion core to the next.

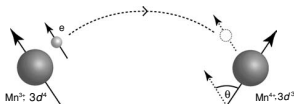
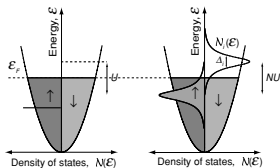


Figure 5.11

The Anderson impurity model. Local density of states for a magnetic impurity in a metal. On the left is shown the case where there is no mixing of the wave functions of the impurity level with the conduction electrons, on the right is the result of such hybridization.



Another common double-exchange pair is Fe^{3+} and Fe^{2+} , which are d^5 and d^6 ions respectively. The d^5 configuration is a half-filled $\uparrow$ shell, and the sixth d electron occupies the bottom of a $t_{2g}^{\downarrow}$ band when the ion is octahedrally coordinated by oxygen where it can hop directly from one d^5 core to another.

5.3 Band magnetism

5.3.1 Magnetic impurities in nonmagnetic metals

The above discussion of exchange between localized moments and conduction electrons in a metal begs the question of whether a magnetic impurity can really retain its moment when diluted in a nonmagnetic matrix. Does a single atom of cobalt, for example, still have a moment when it is diluted in copper? The magnetic impurity problem engrossed the magnetism community in the 1960s and 1970s. The $3d$ electrons of cobalt will hybridize with the $4s$ electrons of copper, broadening the local atomic level into a Lorentzian-like feature in the density of states. Figure 5.11 shows the energy level of a singly occupied d orbital before hybridization, with the doubly occupied orbital higher in energy by the on-site Coulomb repulsion energy U . Hybridization with the conduction band states broadens the impurity level, giving it a width Δ_i . Anderson showed that a moment, albeit one reduced by hybridization, is stable provided $U > \Delta_i$. Further broadening of the local density of states destroys the moment



P. W. Anderson 1923–.

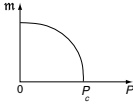
completely. The broadening is more effective for p -metals than for s -metals, because more electrons are available to hybridize with the impurity d -orbitals. Cobalt keeps its moment in copper, but loses it in aluminium.

The number of unpaired impurity electrons $N = N^+ - N^-$ is

$$N = v(\varepsilon_F^+) - v(\varepsilon_F^-), \quad (5.31)$$

where $\varepsilon_F^\pm = \varepsilon_F \pm \frac{1}{2}NU$ and $v(\varepsilon)$, is the integral of the impurity density of states, $\int_0^\varepsilon \mathcal{N}_i(\varepsilon')d\varepsilon'$. Expanding this expression as a power series for small N , we find $N = NU\mathcal{N}_i(\varepsilon_F) + \frac{1}{24}(NU)^3\mathcal{N}_i''(\varepsilon_F)$, where the second derivative $\mathcal{N}_i''(\varepsilon) = d^2\mathcal{N}_i(\varepsilon)/d\varepsilon^2$ is negative. Hence $N^2 = [24(1 - U\mathcal{N}_i(\varepsilon_F))/\mathcal{N}_i''(\varepsilon_F)U^3]$. A moment will form spontaneously at the impurity provided

$$UN_i(\varepsilon_F) > 1. \quad (5.32)$$



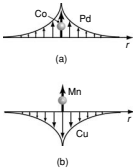
Destruction of an impurity magnetic moment at a critical pressure P_c in the Anderson model.

Since $\mathcal{N}_i(\varepsilon_F)$ is approximately $1/\Delta_i$, we find the Anderson criterion $U \gtrsim \Delta_i$ for magnetism of the impurity. It may be compared with the Stoner criterion for ferromagnetism (5.18). Strong correlations favour magnetism, strong mixing destroys it. If $\mathcal{N}_i(\varepsilon_F)$ varies *smoothly* with some parameter x such as pressure or concentration, then the magnetic moment on the impurity just below the critical value x_c where it disappears will vary as $(x - x_c)^{\frac{1}{2}}$.

The existence of a moment on an atomic site in an alloy may depend sensitively on the local environment. For example, Fe carries a moment when dilute in Mo, but not in Nb. The Fe–Nb hybridization is more effective than Fe–Mo hybridization at broadening the local iron density of states. Iron impurities in $\text{Nb}_{1-x}\text{Mo}_x$ alloys are nonmagnetic when surrounded by less than seven Mo atoms and magnetic when there are seven or more Mo nearest neighbours. In alloys with $x \simeq 0.6$, magnetic and nonmagnetic iron impurities coexist on different sites with different atomic environments. The model where magnetic moments in alloys are governed by the local chemical environment is known as the Jaccarino–Walker model.

Granted a local moment, the s – d Hamiltonian, also known as the Kondo Hamiltonian when $\mathcal{J}_{sd}$ is negative, is written as

$$\mathcal{H} = \sum_{i,j} t_{ij} c_i^\dagger c_j - \sum_{k,l} \mathcal{J}_{sd} \mathbf{S}_k \cdot \mathbf{S}_l. \quad (5.33)$$



(a) A giant moment and (b) a Kondo singlet showing local spin polarization of the host conduction band.

Possible consequences of the interaction between the magnetic impurity and the conduction electrons are the formation of a giant moment when the s – d exchange is ferromagnetic, $\mathcal{J}_{sd} > 0$, or the Kondo effect when the s – d exchange is antiferromagnetic, $\mathcal{J}_{sd} < 0$. The giant moment is due to a cloud of positively polarized electron density surrounding the impurity site. When the paramagnetic susceptibility of the host (5.17) is enhanced beyond the Pauli susceptibility expected from the bare density of states yet not quite sufficiently to meet the Stoner criterion, the dressed local moment can be very large. Cobalt impurities in a palladium host have associated moments of several tens

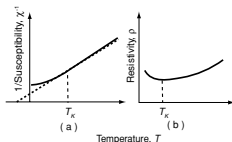
Table 5.3. Kondo temperatures (in kelvin). The host metal is indicated in bold type.

	Cr	Mn	Fe	Co	Ni
Cu	1.0	0.01	25	2000	5000
Ag	.02	.04	3		
Au	0.01	0.01	0.3	200	
Zn	3	1.0	90		
Al	1200	530	5000		

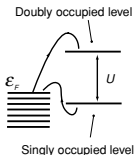
From D. L. Wohlleben and B. R. Coles in *Magnetism 5*, (H. Suhl, editor), New York: Academic Press, 1973.

Figure 5.12

Experimental signs of the Kondo effect: (a) inverse susceptibility of a Kondo alloy and (b) the temperature dependence of the resistivity.



Jun Kondo 1930-.



Kondo scattering. A singlet state is formed between the impurity spin and the conduction electrons.

of Bohr magnetons. There is a threshold beyond which the entire matrix turns ferromagnetic; for Co in Pd, the threshold is only 1.5 at% (see Fig. 10.13(c)).

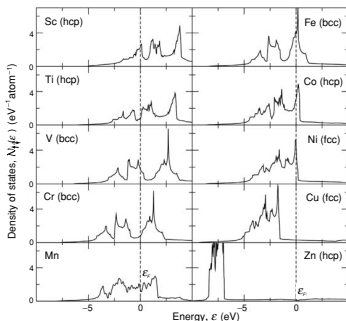
When the exchange coupling between the impurity moment and the conduction electrons $\mathcal{J}_{sd}$ is negative, there is a possibility of forming a nonmagnetic spin singlet state from the impurity and the surrounding cloud of negatively polarized conduction electrons. A good example is iron in copper. The susceptibility is ambiguous; it shows Curie–Weiss temperature dependence (5.7) with negative θ_p above a certain temperature T_K , known as the Kondo temperature, but it becomes temperature-independent below T_K , when the impurity forms a nonmagnetic singlet state with the conduction electrons of the host, Fig. 5.12. According to the system, the Kondo temperature can lie anywhere in the range 1–1000 K. Some values are given in Table 5.3. Another symptom of the Kondo effect is a shallow minimum in the resistivity near T_K , because the Kondo singlets provide an additional channel for scattering conduction electrons. The Kondo temperature is

$$T_K \approx (\Delta_i / k_B) \exp[\Delta_i / 2\mathcal{J}_{sd}]$$

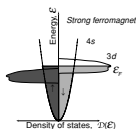
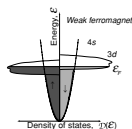
and the excess resistivity due to the Kondo scattering varies logarithmically with temperature; it is proportional to $(\mathcal{J}_{sd}^2 / \Delta_i) S(S+1)[1 + (2\mathcal{J}_{sd} / \Delta_i) \ln(\Delta_i / k_B T)]$.

Figure 5.13

Densities of states for some metallic elements in the paramagnetic state. Calculations by courtesy of Chaitania Das.



5.3.2 Ferromagnetic metals



Schematic densities of states for a strong and a weak ferromagnet. The $3d^1$ -band is full for the strong ferromagnet.

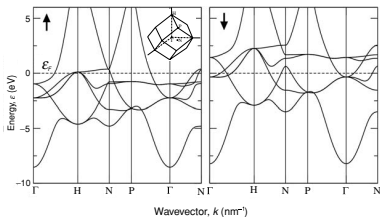
The calculated densities of states for some paramagnetic metals are illustrated in Fig. 5.13. A highly structured $3d$ -band is superposed on a much broader band of $4s$ character. The structure of the d -bands reflects the crystal-field splitting of the t_{2g} - and e_g -bands in 8-fold or 12-fold coordination (Fig. 4.12) in the body-centred cubic (bcc) or face-centred cubic (fcc) structures, as well as the bonding/antibonding splitting between the states near the bottom or the top of the bands, and singularities that appear when the bands cross the boundary of the Brillouin zone.

The band diagrams show the dispersion relations $\varepsilon(k)$ of the five d -bands along different directions in k -space in the first Brillouin zone. The Brillouin zone is a primitive unit cell of reciprocal space defined by the Wigner–Seitz procedure, which involves forming the perpendicular bisector planes of the vectors from the origin to neighbouring reciprocal lattice points. The example of iron metal is illustrated in Fig. 5.14. The spin-up and spin-down bands are shown on separate panels. The broad parabolic free-electron-like s -bands, starting at -4 V are barely spin polarized; they hybridize with the d states between -3 and 2 eV for spin-down. The flatter, spin-up d -bands are filled. Two spin-down d bands lie mainly above Fermi level.

In the Stoner picture of metallic ferromagnetism, the bands split spontaneously provided the criterion (5.18) is satisfied. If the splitting is sufficient

Figure 5.14

Spin-polarized energy bands of ferromagnetic α -iron for $\uparrow$ and $\downarrow$ electrons. The majority spin bands are plotted on the left, and the minority spin bands on the right. The points in the Brillouin zone are marked on the insert. The calculated spin moment is $2.2 \mu_B$. (Calculations courtesy of Chaitanya Das.)



Edmund Stoner,
1899–1968.

to push the $\uparrow$ d -subband completely below ε_F we have a strong ferromagnet, otherwise we have a weak ferromagnet. Among the ferromagnetic elements, Fe is a weak ferromagnet, but Co and Ni are strong (despite their atomic moments being less than that of iron). In each case there are approximately 0.6 electrons at the bottom of an unsplit sp -band. The $3d$ levels lie above the bottom of the $4s$ -band thanks to the term in the Schrödinger equation for the multielectron atom (4.7) which is identified with orbital kinetic energy. The spin moments of Ni and Co are $0.6 \mu_B$ and $1.6 \mu_B$, respectively. Co has a residual unquenched orbital moment of $0.14 \mu_B$. But that of the other ferromagnets is smaller. Iron would have a spin moment of $2.6 \mu_B$ if it were a strong ferromagnet. In fact, its moment is $2.2 \mu_B$. The calculated spin-split densities of states for Fe, Co and Ni are shown in Fig. 5.15.

The different filling of $\uparrow$ and $\downarrow$ bands leads to different Fermi surfaces for $\uparrow$ and $\downarrow$ electrons. They are illustrated for Fe, Co and Ni in Fig. 5.16. The majority spin surfaces for Co and Ni are quite small and roughly spherical because they contain only electrons with predominantly $4s$ character, whereas Fe has a larger $\uparrow$ Fermi surface. All three have large $\downarrow$ Fermi surfaces.

Table 5.4 summarizes the most important properties of the ferromagnetic $3d$ metals.

Stoner calculated the magnetization as a function of temperature in the free-electron model. His calculation gave an unrealistically high Curie point, $k_B T_C \approx \varepsilon_F$ because the only effect of temperature he considered was the smearing of the Fermi–Dirac occupancy function (3.45) which decreases the density of states near ε_F when $k_B T_C \approx \varepsilon_F$. There should be no band splitting and no moment above T_C . The temperature dependence of the susceptibility above T_C is that given by (3.46). However, in most metallic ferromagnets, T_C is at least an order of magnitude less than predicted by the Stoner theory, and there is a substantial Curie–Weiss-like variation of the susceptibility above T_C . On-site electronic correlations sustain an atomic-scale moment which does not

Figure 5.15

Densities of states for some elements in the ferromagnetic state. Fe is a weak ferromagnet, Co and Ni are strong. Results for γ -Fe with different lattice parameters illustrate the sensitivity of the Fe moment to lattice parameter in a dense-packed structure. (Calculations courtesy of Ivan Rungger.)

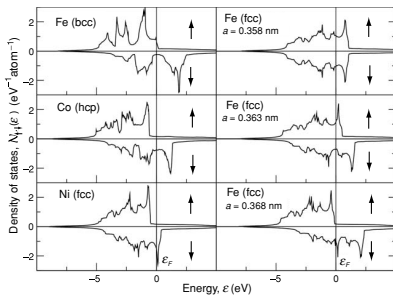


Figure 5.16

The Fermi surfaces of Fe, Co and Ni for $\uparrow$ and $\downarrow$ electrons.

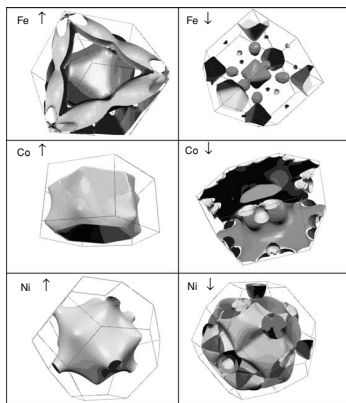


Table 5.4. Intrinsic properties of the ferromagnetic 3d elements at room temperature

	T_C (K)	d (kg m ⁻³)	σ_s (A m ² kg ⁻¹)	M_s (kA m ⁻¹)	J_s (T)	m (spin/orbit) (μ_B)	$N_{\uparrow, \downarrow}(\epsilon_F)$ (eV ⁻¹)	$\mathcal{I}$ (eV)	K_1 (J m ⁻³)	λ_s (10 ⁻⁶)	D_{sw} (10 ⁻⁴⁰ J m ²)	
Fe	1044	7874	217	1710	2.15	2.17 (2.09/0.08)	1.54	0.93	48	-7	2.08	4.5
Co	1360(ϵ)	8920	162(ϵ)	1440	1.81	1.71 (1.57/0.14)	1.72	0.99	410	-60	2.17	8.0
Ni	628	8902	54.8	488	0.61	0.58 (0.53/0.05)	2.02	1.01	-5	-35	2.18	6.3

Table 5.5. Moments in metallic ferromagnets

		m_{eff}	m_0	T_c
Ni	Strong ferromagnet	1.0	0.6	628
ZrZn ₂	Weak itinerant ferromagnet	1.8	0.2	25
CrO ₂	Half-metal	2.4	2.0	396

disappear at T_C , but becomes disordered in much the same way as it does for the local-moment paramagnet (§4.3) or ferromagnet (§5.1.1). The moment is progressively destroyed by thermal fluctuations when $T \gg T_C$.

Whenever a local moment is disordered but stable in temperature, the effective moment m_{eff} deduced from the susceptibility above T_C using (5.7) should be consistent with the zero-temperature ferromagnetic moment m_0 in the sense of Table 5.5. For metals with nonintegral numbers of unpaired electrons per atom, we can define an effective spin S^* by $2\mu_B S^* = m_0$, and a corresponding effective moment m_{eff} as $2\sqrt{S^*(S^* + 1)}\mu_B$. The Stoner model applies best to some very weak itinerant ferromagnets with $S^* \ll \frac{1}{2}$, such as ZrZn₂, an intermetallic compound of two nonmagnetic elements which exhibits a small ferromagnetic moment and a low Fermi energy. For a weak itinerant ferromagnet, m_{eff} is even larger than expected from this formula. The effect of temperature is not just to destroy the long-range intersite atomic correlations, but also to eliminate progressively the on-site Hund's rule correlations that sustain a local moment. The susceptibility therefore falls more rapidly with increasing temperature than predicted by the Curie–Weiss law.

The rigid-band model envisages a fixed, spin-split density of states for the ferromagnetic 3d elements and their alloys, which is filled up with the necessary number of electrons as if they were water being poured into a jug. The jugs for bcc and fcc metals are differently shaped. Ignoring the small contribution of the 4s-band, the average moment per atom is $\langle m \rangle \approx (N_{3d}^{\uparrow} - N_{3d}^{\downarrow})\mu_B$. The total number of 3d electrons is $N_{3d} = N_{3d}^{\uparrow} + N_{3d}^{\downarrow}$, where $N_{3d}^{\uparrow} = 5$ for the strong ferromagnets. Hence

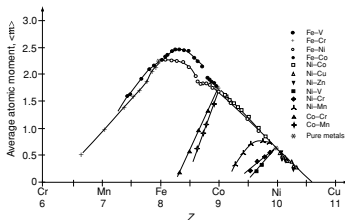
$$\langle m \rangle \approx (10 - N_{3d})\mu_B. \quad (5.34)$$

This relation applies to any strong ferromagnet regardless of the details of the density of states.

The rigid-band picture is oversimplified. Nevertheless, the model has merit. In Cu_xNi_{1-x} alloys, for example, each Cu atom brings an extra electron. The

Figure 5.17

The Slater–Pauling curve. The average atomic moment is plotted against the number of valence ($3d + 4s$) electrons.



d -band of Ni has 0.6 holes and the ferromagnetism of $\text{Cu}_x\text{Ni}_{1-x}$ disappears when the d -band is full up at $x=0.6$, as predicted.

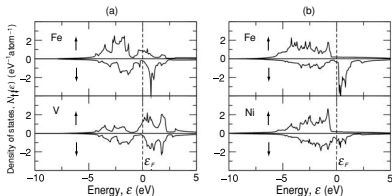
Bonding states with delocalized singlet-like wave functions are found near the bottom of a metallic band and antibonding states with more localized triplet-like states are near the top. Since the total wave function must be antisymmetric, this helps to explain why $3d$ elements near the end of the series tend to be ferromagnetic, while those at the beginning of the series are not. The binding energy of $3d$ electrons increases by about 5 eV across the series, which is comparable to the $3d$ bandwidth. The band narrowing resulting from the increased nuclear charge is sufficient to offset the broadening resulting from reduced metal–metal distances as we move across the $3d$ series.

The famous Slater–Pauling curve, Fig. 5.17, is a plot of the magnetic moment per atom for binary alloys of $3d$ elements plotted against Z , the total number of $3d$ and $4s$ electrons per atom. There is inevitably some mixture of $4p$ character in the $4s$ -band. The alloys on the right-hand side of Fig. 5.17 are strong ferromagnets. The slope of the branch on the right is -1 . The multiple branches with slope ≈ 1 , as expected for rigid bands, are for alloys of late $3d$ elements with early $3d$ elements for which the $3d$ -states lie well above the Fermi level of the ferromagnetic host. The assumption of a common band in the rigid-band model really applies only when the charge difference of the constituent atoms is small, $\Delta Z \lesssim 2$. Otherwise a split band with a joint density of states reflects the densities of states of the constituents. The partial densities of states of FeV and FeNi₃, for example, are compared in Fig. 5.18.

The magnetic valence model is a more general formulation of these ideas which allows us to estimate the average atomic moment per atom of any alloy of a $3d$ element, provided it is a strong ferromagnet. The valence of an atom is given by $Z = N^\uparrow + N^\downarrow$, where $N^\uparrow$ and $N^\downarrow$ are the numbers of $\uparrow$ and $\downarrow$ valence electrons per atom. The magnetic moment is given by $m = (N^\uparrow - N^\downarrow)\mu_B = (2N^\uparrow - Z)\mu_B$. Now the value of $N_d^\uparrow$ is exactly 5 for strong ferromagnetic

Figure 5.18

Partial densities of states of
(a) FeV and (b) FeNi₃.



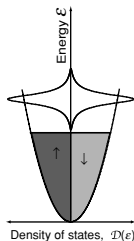
elements, and 0 for main group elements which have no d electrons. The magnetic valence of an element Z_m , defined as

$$Z_m = 2N_d^\uparrow - Z, \quad (5.35)$$

is an integer. Its moment is $m = (Z_m + 2N_s^\uparrow)\mu_B$, where $2N_s^\uparrow \approx 0.6 - 0.7$ is the number of electrons in the unpolarized $4s$ p -band. The average moment per atom in an alloy is obtained by replacing Z_m by its weighted average value over all atoms present in the alloy:

$$\langle m \rangle = (\langle Z_m \rangle + 2N_s^\uparrow)\mu_B. \quad (5.36)$$

In this way it is possible to estimate the magnetization of any strong ferromagnetic alloy based on iron, cobalt or nickel. Some magnetic valences are $Z_m = -3$ for B, Y, La and all rare-earths, -4 for C, Si, Ti, -5 for V, P, -6 for Cr, but 2 for Fe, 1 for Co and 0 for Ni. Taking YFe₂ as an example, the average moment is $[\frac{1}{3}(-3 + 0.6) + \frac{2}{3}(2 + 0.6)] = 0.93 \mu_B/\text{atom}$ or $2.8 \mu_B/(\text{formula unit, fu})$. We can consider that the yttrium has reduced the iron moment from $2.2 \mu_B$ to $1.4 \mu_B$. Adding more yttrium to the alloy will eventually destroy the magnetism entirely (Exercise 5.7(c)). Moments per atom for rare-earth-iron alloys are shown in Fig. 5.19.



A nonmagnetic virtual
bound state.

5.3.3 Impurities in ferromagnets

The converse of the problem considered in §5.3.1, the behaviour of a single impurity atom in a ferromagnetic host, is also interesting. If the impurity is a much lighter $3d$ element than the host, like V in Ni (Fig. 5.20) its d levels lie above the Fermi level in the $4s$ conduction band. If V_{kd} is the hopping integral from the d level to the conduction band, the level acquires a width

$$\Delta_d = \pi \mathcal{N}_{4s}(\epsilon_F) V_{kd}^2, \quad (5.37)$$

Figure 5.19

Variation of the magnetic moment per atom as a function of magnetic valence for some rare-earth iron alloys.

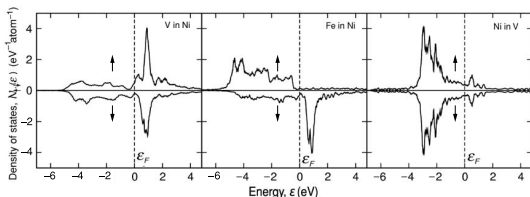
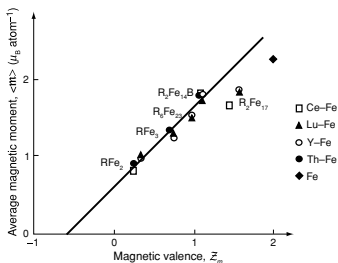


Figure 5.20

The local density of states for V in Ni, Fe in Ni and Ni in V. (Calculations by courtesy of Nadjib Baadji).

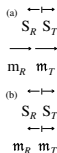
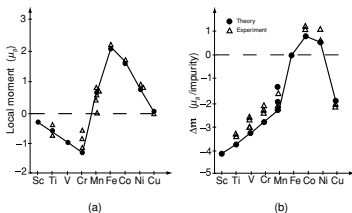
which may be of order 1 eV. The impurity level is then known as a virtual bound state. The width is inversely related to the time an electron dwells on the impurity site.

When the virtual bound state lies entirely above ϵ_F , the 3d impurity electrons are emptied into the 3d-band of the host. If the host 3d[↑]-band is full, there will be a moment reduction of N_{3d}^i Bohr magnetons, where N_{3d}^i is the number of impurity 3d electrons. In addition, the moment of one host atom is suppressed at the site of the substitution. For example, when a V impurity ($Z = 5$, $N_{3d}^i \approx 4$) is substituted in a Ni host, the moment reduction is drastic, $4 + 0.6 = 4.6 \mu_B/\text{V}$.

There will inevitably be some hybridization of the impurity and host 3d-states, which will be more effective for the host 3d[↓] electrons, because they lie closer to the Fermi level. A light 3d element will therefore acquire a small

Figure 5.21

(a) Local moment carried by 3d impurities in iron, and (b) change of moment per impurity atom substituted into iron (O'Handley, 1999).



Coupling of spins and alignment of moments in R-T alloys, T = Fe, Co, Ni (a) R = light rare-earth, (b) R = heavy rare-earth.

negative moment in a heavy 3d host. These trends are illustrated for impurities in iron in Fig. 5.21.

This is an example of the more general rule that *exchange coupling between atoms with d shells that are more than half full with atoms whose d shell is less than half full is antiferromagnetic*. The rare-earths in this context should be considered as light d elements because their atomic configuration is $4f^n 5d^1 6s^2$. There is therefore antiparallel coupling of the *spin* moments of the ferromagnetic 3d elements T = Fe, Co and Ni, and the *spin* moment of a rare-earth. When the 4f shell is half-filled, or more, this leads to *antiparallel* coupling of the atomic moments in R-T alloys with R = Gd–Yb. However, in light rare-earth metals where the moment is mainly orbital in character, and directed opposite to the spin moment according to Hund's third rule, the R and T moments are *parallel*, even though the spins are antiparallel. Many examples of R-T alloys are presented in Chapter 11.

5.3.4 Half-metals

These oddly named materials are ferromagnets with electrons of only one spin polarization at the Fermi level. Cobalt and nickel are *not* half-metals because of the presence of the 4s electrons at ϵ_F which are not fully spin-polarized. Indeed, no ferromagnetic element is a half-metal. It is necessary to form a compound where the 4s electrons can be removed from the vicinity of the Fermi energy by charge transfer or hybridization. Examples include the oxide CrO_2 and the ordered intermetallic compound MnNiSb , which are both discussed in Chapter 11. The characteristic feature of a half-metal is a spin gap in the $\uparrow$ or $\downarrow$ density of states at ϵ_F . Furthermore, the spin moment per formula unit in a stoichiometric half-metallic compound is an *integral number of Bohr magnetons*. This is because there are an integral number $N^\uparrow + N^\downarrow$ of electrons per formula unit,

and the band with the spin gap must contain an integral number of $N^\uparrow$ (or $N^\downarrow$) electrons, hence $N^\uparrow - N^\downarrow$ is also an integer.

Spin-orbit interaction tends to destroy half-metallicity by mixing $\uparrow$ and $\downarrow$ states, as explained in §5.6.4.

5.3.5 The two-electron model

Further insight into the physical interactions of importance in metals is provided by a diatomic two-electron model. Although highly simplified, the model contains most of the ingredients of the physics of the many-electron problem, except orbital degeneracy. Important quantities are the on-site Coulomb repulsion U , the transfer or 'hopping' integral t which gives rise to the bandwidth W and the direct exchange $\mathcal{J}_d$.

The Hamiltonian $\mathcal{H}(\mathbf{r}_1, \mathbf{r}_2)$ is that in the Schrödinger equation of (5.20), with an additional term $e^2/4\epsilon_0 |\mathbf{r}_1 - \mathbf{r}_2|$ to take account of the Coulomb interaction of the two electrons with each other. The spatially symmetric and antisymmetric wave functions

$$\phi_s = (1/\sqrt{2})(\psi_1 + \psi_2), \quad \phi_a = (1/\sqrt{2})(\psi_1 - \psi_2), \quad (5.21)$$

may be regarded as embryonic Bloch functions (electron waves) for the metal with $k = 0$ and $k = \pi/d$, where d is the interatomic spacing. We can replace ϕ_s and ϕ_a by embryonic Wannier functions which are mostly localized on the left and right atoms:

$$\phi_l = (1/N)(\psi_1 + a\psi_2), \quad \phi_r = (1/N)(a\psi_1 + \psi_2), \quad (5.38)$$

with

$$a = \frac{-1 + \sqrt{1 - \mathcal{S}^2}}{\mathcal{S}},$$

where $\mathcal{S}$ is the overlap integral $\int \psi_1^*(r)\psi_2(r)d^3r$ and N is a normalization factor. These Wannier functions, Fig. 5.22, differ from ψ_1 and ψ_2 , the eigenfunctions of the one-electron problem, in that they are supposed to be orthogonal $\int \phi_l^*(r)\phi_r(r)d^3r = 0$.

There are now four possible two-electron wave functions $\Psi_i(\mathbf{r}, \mathbf{r}')$:

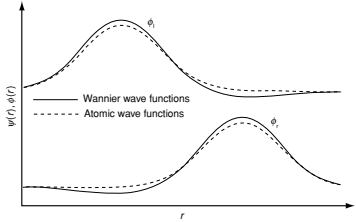
$$\begin{aligned} \Psi_1 &= \phi_l(\mathbf{r})\phi_l(\mathbf{r}'); & \Psi_2 &= \phi_l(\mathbf{r})\phi_r(\mathbf{r}'); \\ \Psi_3 &= \phi_r(\mathbf{r})\phi_l(\mathbf{r}'); & \Psi_4 &= \phi_r(\mathbf{r})\phi_r(\mathbf{r}'). \end{aligned}$$

The functions Ψ_1 and Ψ_4 represent doubly occupied states. The interaction matrix is

$$\begin{pmatrix} U & t & t & \mathcal{J}_d \\ t & 0 & \mathcal{J}_d & t \\ t & \mathcal{J}_d & 0 & t \\ \mathcal{J}_d & t & t & U \end{pmatrix}. \quad (5.39)$$

Figure 5.22

Wannier functions and atomic wave functions for the diatomic molecule.



The Coulomb interaction U is the energy penalty when two electrons are put into the same orbital. It is several electron volts:

$$U = \int \phi_i^*(\mathbf{r})\phi_i^*(\mathbf{r}')\mathcal{H}(\mathbf{r}, \mathbf{r}')\phi_i(\mathbf{r})\phi_i(\mathbf{r}')d^3r d^3r'.$$

The transfer or hopping integral t is also positive, and is $\lesssim 1$ eV. It represents the bandwidth. More generally, in the tight-binding approximation, the bandwidth is $2Z_n t$, where Z_n is the number of nearest neighbours.

$$t \approx \int \phi_R^*(\mathbf{r})\phi_L^*(\mathbf{r}')\mathcal{H}(\mathbf{r}, \mathbf{r}')\phi_L(\mathbf{r})\phi_R(\mathbf{r}')d^3r d^3r'$$

The direct exchange between doubly occupied sites is smaller, and of order 0.1 eV:

$$\mathcal{J}_d = \int \phi_i^*(\mathbf{r})\phi_i^*(\mathbf{r}')\mathcal{H}(\mathbf{r}, \mathbf{r}')\phi_r(\mathbf{r})\phi_r(\mathbf{r}')d^3r d^3r'.$$

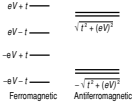
The interaction matrix (5.39) can be diagonalized directly. Two doubly occupied states have eigenvalues of order U , and are therefore neglected. The other states, which are much lower in energy, are: (i) a delocalized ferromagnetic state (the spatial part of the wave function is antisymmetric) with eigenvalue $\varepsilon_{FM} = -\mathcal{J}_d$

$$\Psi_{FM} = (1/\sqrt{2})[\phi_l(\mathbf{r})\phi_r(\mathbf{r}') - \phi_r(\mathbf{r})\phi_l(\mathbf{r}')];$$

and (ii) an antiferromagnetic state (the spatial part of the wave function is symmetric)

$$\begin{aligned} \Psi_{AF} = & (\sin \chi / \sqrt{2})[\phi_l(\mathbf{r})\phi_l(\mathbf{r}') + \phi_r(\mathbf{r})\phi_r(\mathbf{r}')] \\ & + (\cos \chi / \sqrt{2})[\phi_l(\mathbf{r})\phi_r(\mathbf{r}') + \phi_r(\mathbf{r})\phi_l(\mathbf{r}')], \end{aligned}$$

where $\tan \chi = 4t/U$. The associated energy is $\varepsilon_{AF} = U/2 + \mathcal{J}_d - \sqrt{4t^2 + U^2/4}$.



Energy levels of the ferromagnetic and antiferromagnetic states for the two-atom/four-state problem.

The effective exchange is $\mathcal{J}_{\text{eff}} = \frac{1}{2}(\varepsilon_{AF} - \varepsilon_{FM})$,

$$\mathcal{J}_{\text{eff}} = \mathcal{J}_d + U/4 - \sqrt{(t^2 + U^2/16)}. \quad (5.40)$$

$\mathcal{J}_{\text{eff}} > 0$ indicates a ‘ferromagnetic’ ground state and $\mathcal{J}_{\text{eff}} < 0$ indicates an ‘antiferromagnetic’ ground state. Direct exchange favours ferromagnetism, but strong interatomic hopping t favours antiferromagnetism. When $U \gg t$, and $\mathcal{J}_d = 0$, the exchange is antiferromagnetic, as illustrated in §5.2.2:

$$\mathcal{J}_{\text{eff}} = -2t^2/U. \quad (5.41)$$

Our rudimentary model can also illustrate how exchange depends on band filling. We consider the ferromagnetic and the antiferromagnetic states, for which the one-electron Hamiltonians are,

$$\mathcal{H}_F = \begin{bmatrix} \pm eV & t \\ t & \pm eV \end{bmatrix},$$

$$\mathcal{H}_{AF} = \begin{bmatrix} \pm eV & t \\ t & \mp eV \end{bmatrix},$$

where V is the local exchange potential experienced by an electron on site 1 or 2 and t is the interatomic hopping integral. When $t = 0$, there is one-electron exchange splitting of the states. Diagonalizing the matrices to find the eigenvalues involves solving the determinant $|\mathcal{H} - \lambda \mathbf{I}| = 0$. For the ferromagnetic state, the energy levels are $\pm eV + t$ and $\pm eV - t$, whereas for the antiferromagnetic state, they are doubly degenerate $\pm \sqrt{t^2 + (eV)^2}$. It can be seen that a single electron or three electrons (quarter-filled or three-quarters-filled band) go into a ferromagnetic state, but two electrons (half-filled band) prefer the antiferromagnetic state.

5.3.6 The Hubbard model

A famous model Hamiltonian which represents electron correlation in the tight-binding model for an array of one-electron atoms is

$$\mathcal{H} = - \sum_{i,j} t c_i^\dagger c_j + U \sum_i N_i^\uparrow N_i^\downarrow,$$

where $N_i^{\uparrow,\downarrow}$ are the numbers of spin-up and spin-down electrons, respectively, on the i th atom. The first term is the transfer term that creates the band of width $W = 2Zt$; the second is the Coulomb energy penalty involved in placing two electrons on the same atom.

Electrons are localized when $U/W > 1$ because there are then no states available to accommodate the double-occupancy charge fluctuations that are indispensable for electronic conduction. Compounds with an integral number of electrons per atom which satisfy this condition are known as Mott-Hubbard insulators.

The second term can be rewritten using

$$UN_i^\dagger N_i^\downarrow = U[(N^\uparrow + N^\downarrow)^2/4 - (N^\uparrow - N^\downarrow)^2/4].$$

The Stoner interaction $-(\mathcal{I}/4)(N^\uparrow - N^\downarrow)^2$ is thereby identified as the spin-dependent part of the on-site Coulomb interaction; hence $\mathcal{I} \approx U$. In the Hubbard model, the on-site correlations create a magnetic moment, and hopping between adjacent nondegenerate singly occupied orbitals provides an antiferromagnetic interaction.

A variant of the Hubbard model is the t - $\mathcal{J}$ model, where the second term is replaced by $-2\mathcal{J} \sum_{i>j} \mathbf{S}_i \cdot \mathbf{S}_j$ with $\mathcal{J} = -2t^2/U$.

5.3.7 Electronic structure calculations



The Born-Oppenheimer approximation: the electrons (marked with arrows) move in a background of frozen nuclei (open circles).

A solid is a system of N electrons at positions $\{\mathbf{r}_i\}$ and N' nuclei, usually centred on a periodic lattice $\{\mathbf{R}_I\}$. Inner electrons occupy tightly bound, localized core orbitals around the nuclei. Outer orbitals with binding energies of a few electron volts or less are the home of the valence and conduction electrons, which determine the electronic character of the solid, be it metal, semiconductor, insulator, ferromagnet, antiferromagnet, superconductor, ... The instantaneous velocity of these outer electrons is of order the Fermi velocity, $v_F \approx 10^6 \text{ m s}^{-1}$. Ion cores vibrate at phonon frequencies that are of order 10^{14} Hz , with an amplitude of about 10 pm , which means that their velocity is of order 10^3 m s^{-1} . We are therefore justified in thinking that the electron sees the potential of a set of nuclei instantaneously frozen in position – the Born-Oppenheimer approximation. Furthermore, we will ignore the atomic displacements, which lead to electron scattering, and assume that the ion cores are localized at the lattice sites. The electrons experience Coulomb interactions with the nuclei, and with each other. The Hamiltonian, with factors of $\frac{1}{2}$ to avoid double counting, takes the form

$$\mathcal{H} = - \sum_i \frac{\hbar^2 \nabla_i^2}{2m_e} - \sum_{i,I} \frac{Ze^2}{4\pi\epsilon_0 R_{Ii}} + \frac{1}{2} \sum_{i,j} \frac{e^2}{4\pi\epsilon_0 r_{ij}}. \quad (5.42)$$

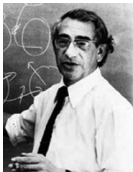
The problem is to solve Schrödinger's equation $\mathcal{H}\Psi = \varepsilon\Psi$, where $\Psi(\{\mathbf{R}_I\}, \{\mathbf{r}_i\})$ is a wave function for the huge number of electrons and nuclei in the system.

An abbreviated notation for $\mathcal{H}$ is

$$\mathcal{H} = \mathcal{T} + \mathcal{V} + \mathcal{U}, \quad (5.43)$$

where $\mathcal{T}$ and $\mathcal{V}$ are the terms corresponding to the one-electron kinetic and potential energy and $\mathcal{U}$ represents the two-electron interactions that capture the complexity of the physics.

Many first-principles methods for solving the many-electron Schrödinger equation use wave functions based on Slater determinants. The idea is to build in the antisymmetry of the wave function under exchange of any two electrons



John Hubbard, 1931–1980

with given space and spin coordinates x, y, z, σ denoted as $\mathbf{x}_i$ and $\mathbf{x}_j$. In the case of just two electrons, for example, $\Psi = \psi_1(\mathbf{x}_1)\psi_2(\mathbf{x}_2)$ is not antisymmetric, but $(1/\sqrt{2})[\psi_1(\mathbf{x}_1)\psi_2(\mathbf{x}_2) - \psi_1(\mathbf{x}_2)\psi_2(\mathbf{x}_1)]$ is a suitable wave function. It can be written in the form of a determinant:

$$\Psi(\mathbf{x}_1, \mathbf{x}_2) = \frac{1}{\sqrt{2}} \begin{vmatrix} \psi_1(\mathbf{x}_1) & \psi_2(\mathbf{x}_1) \\ \psi_1(\mathbf{x}_2) & \psi_2(\mathbf{x}_2) \end{vmatrix}. \quad (5.44)$$

Placing two electrons in the same orbit $\psi_1 = \psi_2$ gives $\Psi(\mathbf{x}_1, \mathbf{x}_2) = 0$, as required by the Pauli principle. Slater generalized this idea to N electrons, writing the wave function as

$$\Psi(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N) = \frac{1}{\sqrt{N!}} \begin{vmatrix} \psi_1(\mathbf{x}_1) & \psi_2(\mathbf{x}_1) & \dots & \psi_N(\mathbf{x}_1) \\ \psi_1(\mathbf{x}_2) & \psi_2(\mathbf{x}_2) & \dots & \psi_N(\mathbf{x}_2) \\ \vdots & \vdots & \ddots & \vdots \\ \psi_1(\mathbf{x}_N) & \psi_2(\mathbf{x}_N) & \dots & \psi_N(\mathbf{x}_N) \end{vmatrix}. \quad (5.45)$$

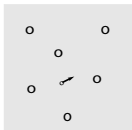
A compact way to denote a Slater determinant is as a ket, $|1, 2, \dots, N\rangle$.

The Hartree-Fock method assumes that the exact N -electron wave function of the system can be approximated as a single Slater determinant. A variational solution is then based on a linear combination of these one-electron wave functions with coefficients chosen to minimize the energy. The method completely neglects electron correlations, but takes perfect account of the exchange. Each electron is surrounded by an exchange hole, from which any other electron with the same spin is excluded.

An alternative approach to Hartree-Fock calculations is density functional theory (DFT), which provides an approximate solution for both exchange and correlation energies. It succeeds in mapping a many-electron problem with $\mathcal{U}$ onto a one-electron problem without $\mathcal{U}$. The theory is based on two theorems, proved by Hohenberg and Kohn in the mid 1960s. The first is that *the density $n(\mathbf{r})$ of a system of N electrons determines all the ground-state electronic properties*. The ground-state wave function Ψ_0 is a unique functional of electron density $\Psi_0[n(\mathbf{r})]$. (A functional is just a function which has another function as its argument.) Other physical properties can be derived from the wave function. In particular, the ground-state energy is

$$\varepsilon_0[n(\mathbf{r})] = \langle \Psi_0 | T + V + \mathcal{U} | \Psi_0 \rangle. \quad (5.46)$$

The second is that *the energy functional $\varepsilon_0[n(\mathbf{r})]$ is lower in energy for the ground-state density $n_0(\mathbf{r})$ than any other state*. The term in the energy which needs to be minimized in (5.46) is the one that depends on $\{R_I\}$, $V[n(\mathbf{r})] = -e \int V(\mathbf{r})n(\mathbf{r})d^3r$. The significance of these theorems is that it is immensely easier to base a calculation on the density, which depends on only three variables, x, y, z , or four when we include the spin σ , than it is on the wave functions of an N -electron system which depend on $4N$ variables. The problem is that the correct density functional is unknown, and it must be arrived at by inspired approximation. Furthermore, the method applies to the ground state,



The Kohn-Sham formalism. Each electron moves independently in an effective potential created by the others.

and it does not give the structure of excited states, although a time-dependent variant of the theory may remedy this defect.

Density functional theory is usually implemented using the Kohn-Sham method, where the problem of strongly interacting electrons moving in the potential of the nuclei is reduced to the more tractable problem of noninteracting electrons in an effective potential which somehow takes care of the inter-electronic Coulomb interaction, as well as exchange and correlation effects. The energy of (5.46) is rewritten as

$$\varepsilon_0[n(\mathbf{r})] = T_s[n(\mathbf{r})] + \varepsilon_V[n(\mathbf{r})], \quad (5.47)$$

where T_s is the noninteracting kinetic energy and ε_V is the total potential energy. The Kohn-Sham equations for this noninteracting system are just a set of effective, single-particle Schrödinger equations

$$\left[-\frac{\hbar^2 \nabla_i^2}{2m_e} + \mathcal{V}_s(\mathbf{r}) \right] \phi_i(\mathbf{r}) = \varepsilon_i \phi_i(\mathbf{r}) \quad (5.48)$$

which yield a set of orbitals ϕ_i which are approximate wave functions for the real system of electrons that reproduce the density of the original many-electron system $n(\mathbf{r}) = \sum_i |\phi_i^2|$. The effective single-particle potential is usually written as

$$\mathcal{V}_s = \mathcal{V} + \frac{1}{2} \int \frac{e^2 n(\mathbf{r}')}{4\pi\epsilon_0 |\mathbf{r} - \mathbf{r}'|} d^3 r' + \mathcal{V}_{xc}[n(\mathbf{r})]. \quad (5.49)$$

The first term is the Coulomb interaction of the electron with the nuclei, the second term is the Hartree term $\mathcal{V}_H$ describing the electron-electron Coulomb repulsion, and the key term is the third one, the exchange-correlation potential, which includes all the many-electron correlations. The Kohn-Sham equations are solved by following an iterative procedure. Taking an initial guess for $[n(\mathbf{r})]$, $\mathcal{V}_s$ is calculated and the equations are solved for $\phi_i(\mathbf{r})$, from which a new density is calculated, and the process is repeated. The local density approximation (LDA) assumes that the exchange-correlation functional $\mathcal{V}_{xc}[n(\mathbf{r})]$ depends only on the density at the point where the function is evaluated $\mathcal{V}_{xc}[n(\mathbf{r})]$. A variant is the general gradient approximation, where $\mathcal{V}_{xc}$ depends also on the density gradient $\mathcal{V}_{xc}[n(\mathbf{r}), \nabla n(\mathbf{r})]$. In systems of more ionic character, a U term can be added to reduce double orbital occupancy.

All this can be generalized to the spin-dependent case. In the local spin density approximation (LSDA) two densities must be taken into account, the scalar electron density $n(\mathbf{r})$ and the vector magnetization density $\mathbf{m}(\mathbf{r}) = \mu_B [n^\uparrow(\mathbf{r}) - n^\downarrow(\mathbf{r})] \mathbf{e}_z$. They are both incorporated in a 2×2 density matrix $\hat{n}(\mathbf{r}) = \frac{1}{2} [n(\mathbf{r}) \hat{\mathbf{I}} + \hat{\boldsymbol{\sigma}} \cdot \mathbf{s}(\mathbf{r})]$, where $\hat{\mathbf{I}}$ is the identity matrix $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $\hat{\boldsymbol{\sigma}}$ are the Pauli spin matrices (3.17) and $\mathbf{s}(\mathbf{r})$ is the local spin density $\mathbf{m}(\mathbf{r})/\mu_B$. For a

collinear spin configuration, the density matrix is diagonal:

$$\hat{n}(\mathbf{r}) = \begin{bmatrix} n^\uparrow(\mathbf{r}) & 0 \\ 0 & n^\downarrow(\mathbf{r}) \end{bmatrix}, \quad (5.50)$$

so that $n(\mathbf{r}) = n^\uparrow(\mathbf{r}) + n^\downarrow(\mathbf{r})$ and $\mathbf{m}(\mathbf{r}) = \mu_B[n^\uparrow(\mathbf{r}) - n^\downarrow(\mathbf{r})]$. The potential matrix is given by $\hat{V} = V\hat{\mathbf{I}} + \mu_B\mathbf{m}(\mathbf{r}) \cdot \mathbf{B}$, where $\mathbf{B}$ is the magnetic field. The spin-polarized version of the Kohn–Sham equations is

$$\left[\left(-\frac{\hbar^2 \nabla^2}{2m_e} + V_H \right) \hat{\mathbf{I}} + V + V_{xc} \right] \begin{bmatrix} \phi_i^\uparrow(\mathbf{r}) \\ \phi_i^\downarrow(\mathbf{r}) \end{bmatrix} = \varepsilon_i \begin{bmatrix} \phi_i^\uparrow(\mathbf{r}) \\ \phi_i^\downarrow(\mathbf{r}) \end{bmatrix}. \quad (5.51)$$

The exchange-correlation matrix depends on both $n(\mathbf{r})$ and $\mathbf{m}(\mathbf{r})$,

$$V_{xc} = V_{xc}[n(\mathbf{r}), \mathbf{m}(\mathbf{r})], \quad (5.52)$$

for which a suitable approximation must be found. Equations (5.51) yield the wave functions $\phi_i^{\uparrow,\downarrow}(\mathbf{r})$ from which the density matrix $\hat{n}(\mathbf{r})$ is deduced. A self-consistent solution is obtained, as in the nonmagnetic case. A number of computer codes are available to do the job. If the density matrix is diagonal, the magnetic structure is collinear, but the general formalism allows for noncollinear structures. DFT is an accurate method for calculating magnetic moments and spin-polarized band structures, especially in metallic systems.

Exponential growth of computer power, and especially multiprocessor computer clusters, has enabled computer simulation to establish itself as a third force, alongside experiment and theory, for investigations in magnetism. Not only in electronic structure calculations, where it is becoming possible to investigate the crystal structure and magnetic order of a new compound without ever actually having to make it in the laboratory, but also in the areas of electronic transport properties and micromagnetism, are computational methods making their mark. Large numbers of atoms, of order 1000 or more, can be handled using current DFT codes, which makes it possible to investigate spin-dependent transport in a molecule, or to study the appearance of a magnetic moment on different types of lattice defects in a solid. It is immensely more convenient to create a specific complex lattice defect on a computer than it is in the laboratory.



The three forces in magnetism: theory, experiment, and simulation. (Courtesy Wiebke Drenckhan).

5.4 Collective excitations

The comparison of magnetization data on nickel with the predictions of molecular field theory for $J = \frac{1}{2}$ in Fig. 5.3 shows discrepancies both at low temperature, and in the vicinity of T_C . Actually the discrepancies are worse than they appear because T_C is used to determine n_W , so the model is constrained to return the right Curie temperature, and the correct value of m_0 .

Experimental methods discussed in Chapter 10, exist to determine the exchange constants directly, so it is possible to make a more telling comparison

Figure 5.23

Spontaneous magnetization and inverse susceptibility as a function of temperature. The molecular field predictions (grey) are compared with typical experimental data (black).

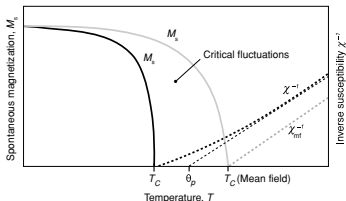
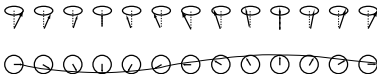


Figure 5.24

Illustration of a spin wave.



between theory and experiment, as indicated in Fig. 5.23. Here it is evident that ferromagnetism is considerably less stable at elevated temperature than molecular field theory would have us believe; it overestimates T_C by as much as a factor of 2, depending on the dimensions and the lattice type. The spontaneous magnetization is diminished at low temperatures by spin-wave excitations. Near T_C the critical fluctuations destroy it.

5.4.1 Spin waves

The total exchange energy in the ferromagnetic ground state is $-2Z\mathcal{J}S^2$ per site, where Z is the number of magnetic nearest neighbours and $\mathcal{J}$ is the nearest-neighbour exchange interaction. The elementary excitations from the ferromagnetic ground state are not, as might be imagined, flips of individual spins that reduce an atomic moment from a state with $M_s = S$ to one with $M_s = (S - 1)$. A single localized spin reversal in an $S = \frac{1}{2}$ chain $\uparrow\uparrow\uparrow\downarrow\uparrow\uparrow\uparrow$ costs $8\mathcal{J}S^2$ or $2\mathcal{J}$ when $S = \frac{1}{2}$, which is twice as large as $k_B T_C$ for the chain treated in the molecular field approximation (5.26); $Z = 2$ for a chain, so $k_B T_C = 2\mathcal{J}ZS(S + 1)/3 = \mathcal{J}$. Such expensive excitations cannot occur at low temperature. Instead, all the atoms share out the spin reversal, with periodic oscillation of their transverse spin orientation. The spin deviations spread over the whole lattice in a propagating spin wave with wave vector $\mathbf{q}$ and energy $\varepsilon_{\mathbf{q}} = \hbar\omega_{\mathbf{q}}$, as illustrated in Fig. 5.24. Spin waves exist as classical excitations, but the extended, quantized spin deviations in solids are known as magnons by analogy with phonons, the quantized lattice waves. Think of spin waves

as oscillations in the relative orientations of spins on a lattice, whereas lattice waves are oscillations of the relative positions of atoms on a lattice.

The relation between the wavevector $q = 2\pi/\lambda$ and frequency ω_q of the spin wave can be calculated classically, or from quantum mechanics. The classical approach considers the spin angular momentum of the atom at site j , $\hbar \mathbf{S}_j$ and equates the torque exerted by the molecular field to the rate of change of angular momentum, thus

$$\hbar \frac{d\mathbf{S}_j}{dt} = \mu_0 g \mu_B \mathbf{S}_j \times \mathbf{H}^j. \quad (5.53)$$

In a chain, the molecular field $\mathbf{H}^j$ at site j is due to the neighbours at sites $j \pm 1$. From (5.25), $\mathbf{H}^j = 2\mathcal{J}(\mathbf{S}_{j-1} + \mathbf{S}_{j+1})/\mu_0 g \mu_B$, hence $\hbar d\mathbf{S}_j/dt = 2\mathcal{J} \mathbf{S}_j \times (\mathbf{S}_{j-1} + \mathbf{S}_{j+1})$. This can be written in Cartesian coordinates:

$$\hbar \frac{dS_j^x}{dt} = 2\mathcal{J}[S_j^y(S_{j-1}^z + S_{j+1}^z) - S_j^z(S_{j-1}^y + S_{j+1}^y)]$$

plus cyclic permutations. For small deviations, we can approximate $S_j^z = S_j = S$ and neglect terms like $S_i^x S_j^y$. Hence

$$\begin{aligned} \hbar \frac{dS_j^x}{dt} &= 2\mathcal{J}S[2S_j^y - S_{j-1}^y - S_{j+1}^y], \\ -\hbar \frac{dS_j^y}{dt} &= 2\mathcal{J}S[2S_j^x - S_{j-1}^x - S_{j+1}^x], \\ \hbar \frac{dS_j^z}{dt} &= 0. \end{aligned} \quad (5.54)$$

Solutions are of the form $S_j^x = uS \exp[i(jqa - \omega_q t)]$, $S_j^y = vS \exp[i(jqa - \omega_q t)]$, where q is the wavevector and a is the interatomic spacing. Substituting back into (5.54) gives $-i\hbar\omega_q u = 4\mathcal{J}S(1 - \cos qa)v$, $i\hbar\omega_q v = 4\mathcal{J}S(1 - \cos qa)u$. Multiplying these results for a one-dimensional chain of isotropic spins, gives

$$\hbar\omega_q = 4\mathcal{J}S(1 - \cos qa). \quad (5.55)$$

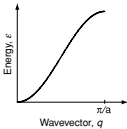
In the limit of small wavevectors, the spin-wave dispersion relation becomes

$$\varepsilon_q \approx D_{sw} q^2, \quad (5.56)$$

where $\varepsilon_q = \hbar\omega_q$ and the spin-wave stiffness parameter is $D_{sw} = 2\mathcal{J}Sa^2$. It takes a vanishingly small energy to create a long-wavelength magnetic excitation. The generalization to a three-dimensional cubic lattice with nearest-neighbour interactions is

$$\hbar\omega_q = 2\mathcal{J}S \left[Z - \sum_{\delta} \cos \mathbf{q} \cdot \boldsymbol{\delta} \right],$$

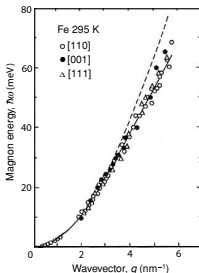
where the sum is over the Z vectors $\boldsymbol{\delta}$ connecting the central atom to its nearest neighbours. The same dispersion relation with $D_{sw} = 2\mathcal{J}Sa_0^2$ is found in any of



The spin-wave dispersion relation for a chain of atoms.

Figure 5.25

The magnon dispersion relations for iron measured in different directions in the unit cell. The dashed line corresponds to $D_{sw} = 4.5 \times 10^{-40} \text{ J m}^2$. (G. Shirane *et al.*, *Journal of Applied Physics*, **39**, 383 (1968))



the three basic cubic lattices, where a_0 is now the lattice parameter. Dispersion of magnons differs from that of phonons, $\varepsilon_q \approx c_0 q$ where c_0 is the velocity of sound, which is linear in the small- q limit. The value of D_{sw} for cobalt, for example, is $8.0 \times 10^{-40} \text{ J m}^2$ (500 meV Å²). It is smaller for other ferromagnets (Table 5.4). The dispersion relation for iron is shown in Fig. 5.25.

Equation (5.55) can be derived quantum mechanically from the Heisenberg Hamiltonian (5.24) where the sum is over nearest-neighbour pairs i, j . The Hamiltonian

$$\mathbf{S}_i \cdot \mathbf{S}_j = S_i^x S_j^x + S_i^y S_j^y + S_i^z S_j^z \quad (5.57)$$

is written in terms of the raising and lowering operators.

$$\mathbf{S}_i \cdot \mathbf{S}_j = S_i^z S_j^z + \frac{1}{2}(S_i^+ S_j^- + S_i^- S_j^+). \quad (5.58)$$

The ground state of the system $|\Phi\rangle$ has all the spins aligned in the z -direction, so that $\mathcal{H}|\Phi\rangle = -2\mathcal{J}(N-1)S^2|\Phi\rangle$. Flipping a spin $\frac{1}{2}$ at site i using S_i^- reduces M_S^i from S to $S-1$; $|i\rangle = S_i^-|\Phi\rangle$ lowers the total spin of the system by 1. However, $|i\rangle$ is not an eigenstate of the Hamiltonian of the chain of spins with nearest-neighbour interactions

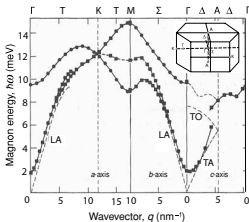
$$\mathcal{H} = -2\mathcal{J} \sum_{i=1}^{N-1} [S_i^z S_{i+1}^z + \frac{1}{2}(S_i^+ S_{i+1}^- + S_i^- S_{i+1}^+)]$$

because $\mathcal{H}|i\rangle = 2\mathcal{J}[-(N-1)S^2 + 2S|i\rangle - S|i+1\rangle - S|i-1\rangle]$. It is necessary to form linear combinations like

$$|q\rangle = \frac{1}{\sqrt{N}} \sum_i e^{iq \cdot \mathbf{r}_i} |i\rangle \quad (5.59)$$

Figure 5.26

Magnon dispersion relation for terbium (J. Jonsen and A. R. Mackintosh, *Rare Earth Magnetism*, Oxford University Press 1991).



This state is a magnon, a spin flip delocalized on the chain, with wavevector q . Then

$$\begin{aligned}\mathcal{H}(\mathbf{q}) &= \frac{2\mathcal{J}}{\sqrt{N}} \sum_{i=1}^{N-1} e^{i\mathbf{q}\cdot\mathbf{r}_i} [-(N-1)S^2 + 2S|i\rangle - S|i+1\rangle - S|i-1\rangle] \\ &= [-2\mathcal{J}(N-1)S^2 + 4\mathcal{J}S(1 - \cos qa)]|\mathbf{q}\rangle.\end{aligned}\quad (5.60)$$

Dropping the first term, which is constant, we have $\varepsilon(\mathbf{q}) = 4\mathcal{J}S(1 - \cos qa)$, as before.

Dispersion relations are best measured by inelastic neutron scattering, which is discussed in Chapter 10. There are multiple magnon branches when the unit cell is noncubic, or if it contains more than one magnetic atom. The energy in a mode of frequency ω_q containing N_q magnons is $(N_q + \frac{1}{2})\hbar\omega_q$. Excitation of magnons is responsible for the fall of magnetization with increasing T . They also contribute to resistivity and magnetic specific heat. By analysing spin-wave dispersion relations measured across the Brillouin zone, it is possible to deduce the exchange interactions $\mathcal{J}(\mathbf{r}_{ij})$ for different atom pairs. Alternatively, the wavevector-dependent exchange $\mathcal{J}(\mathbf{q})$ can be fitted to the data. When the minimum of $\mathcal{J}(\mathbf{q})$ does not fall at $q = 0$, a spatially modulated magnetic structure is stable (§6.3).

Figure 5.26 shows the spin-wave dispersion relation for terbium. There is an energy gap at $q = 0$ due to the single-ion anisotropy of this rare-earth metal (§4.4.4). Excitation of spin waves can be suppressed at very low temperatures by the anisotropy. The energy gap at $q = 0$ is K_1/n , where n is the number of atoms per unit volume. In hexagonal close packed (hcp) cobalt, for example, $n = 9 \times 10^{28} \text{ m}^{-3}$, $K_1 = 500 \text{ kJ m}^{-3}$, the spin-wave gap is 0.4 K.

Magnons behave like bosons; each magnon corresponds to the reversal of one spin $\frac{1}{2}$ over the whole sample, or a change $\Delta M_S = 1$ for the whole system. Hence the average number of quantized spin waves in a mode q is given by the

Table 5.6. Comparison of excitations in solids

Excitation		Dispersion	Specific heat
Electrons	Fermions	$\varepsilon_k \approx (\hbar/2m)k^2$	γT
Phonons	Bosons	$\varepsilon_q \approx c_0 q$	T^3
Magnons (ferromagnetic)	Bosons	$\varepsilon_q \approx D_{fw} q^2$	$T^{3/2}$
Magnons (antiferromagnetic)	Bosons	$\varepsilon_q \approx D_{af} q$	T^3

Bose distribution

$$\langle N_q \rangle = 1/[\exp(\hbar\omega_q/k_B T) - 1].$$

However, the magnon density of states $\mathcal{N}(\omega_q) \propto \omega_q^{-\frac{1}{2}}$, just like that of electrons which have a similar dispersion relation $\varepsilon_k = \hbar^2 k^2/2m$. Dispersion relations and the corresponding low-temperature specific heats are summarized in Table 5.6. It can be shown that the reduction in magnetization at low temperatures due to the excitation of magnons is

$$\Delta M/M_0 = (0.0587/\nu)(k_B T/2S\mathcal{J})^{3/2}. \quad (5.61)$$

This is the Bloch $T^{3/2}$ power law. The integer ν equals 1, 2 or 4 for a simple cubic, bcc or fcc lattice. Specific heat follows the same power law at low temperature. A consequence of spin-wave excitation is that Curie temperatures are much lower than expected from molecular field theory, given the exchange constant $\mathcal{J}$ (Fig. 5.23).

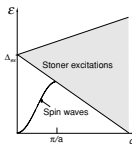
For electrons in ferromagnetic metals, there is an additional scattering process, in addition to scattering from defects, phonons and other electrons. The electron can be inelastically spin-flip scattered, with the creation or annihilation of a magnon ($\omega_q, \mathbf{q}$). This leads to a term in resistivity varying as T^2 .

Our discussion of spin waves has been based on localized spins and Heisenberg exchange coupling, but the idea is more general; any ferromagnetic continuum with exchange stiffness will exhibit spin-wave excitations.

5.4.2 Stoner excitations

Besides spin waves, another type of excitation in a metal can reduce its magnetization. Electrons at the Fermi level can be excited from filled states in the majority-spin band to empty states in the minority-spin band. If the initial state has wavevector $\mathbf{k}$ and the final state has wavevector $\mathbf{k} - \mathbf{q}$, an excitation of wavevector $\mathbf{q}$ is produced. The energy of the excitation is given by

$$\hbar\omega_q = \varepsilon_k - \varepsilon_{k-q} + \Delta_{ex}.$$



Spin waves and Stoner excitations (shaded) in a ferromagnet.

At $q = 0$, these excitations require the full exchange splitting Δ_{ex} , but at finite q there is a broad continuum of Stoner excitations.

5.4.3 Mermin-Wagner theorem

The derivation of the spin-wave dispersion has been based on the existence of a ferromagnetic state in an isotropic chain, or a three-dimensional lattice. The assumptions warrant scrutiny. The number of magnons excited at a temperature T is given by

$$n_m = \int_0^\infty \frac{\mathcal{N}(\omega_q) d\omega_q}{e^{\hbar\omega_q/kT} - 1},$$

where the density of states for magnons $\mathcal{N}(\omega_q)$ in one, two and three dimensions varies as $\omega_q^{-1/2}$, $\omega_q^0 = \text{constant}$ and $\omega_q^{1/2}$, respectively. The argument is similar to that for the electron gas, given in §3.2.5, which has similar dispersion relations. Setting $x = \hbar\omega_q/k_B T$, the integral in three dimensions varies as $(k_B T/\hbar)^{3/2} \int_0^\infty x^{1/2} dx / (e^x - 1)$, whence comes the Bloch $T^{3/2}$ law (5.61). However, the integrals *diverge* at finite temperature in one and two dimensions. The ferromagnetically ordered state should be unstable in dimensions lower than 3. This is the Mermin-Wagner theorem. Magnetic order is possible in the Heisenberg model in three dimensions, but not in one or two. The linear chain, our example of spin-wave dispersion, cannot order except at $T = 0$ K.

The consequences of this theorem are not as catastrophic as they seem at first sight. The divergence is avoided if there is some anisotropy in the system, which creates a gap in the spin-wave spectrum at $q = 0$; the lower limit of integration is then greater than zero and the divergence is avoided. Some anisotropy is always caused by crystal field or dipolar interactions. Two-dimensional ferromagnetic layers do exist in reality, thanks to anisotropy (§8.1).

5.4.4 Critical behaviour

Not only at low temperatures does the mean-field theory fail to account properly for the temperature dependence of the magnetization of a ferromagnet. There is a discrepancy in the critical region, close to T_C where the variation of M with temperature is as $(T - T_C)^\beta$ with $\beta \approx 0.34$, rather than $\frac{1}{2}$ (5.14). If the magnetization is calculated from molecular field theory using the measured exchange parameters $\mathcal{J}(r_{ij})$ derived from spin-wave dispersion relations, the discrepancy in the predicted and measured values of T_C may be as great as 60%. This is partly due to spin waves, but the exchange parameters are renormalized to lower values on increasing temperature because of critical fluctuations. These

remarkable fluctuations, which are self-similar over a wide range of length scales, are discussed in Chapter 6.

5.5 Anisotropy

Magnetic anisotropy means that the ferromagnetic or antiferromagnetic axis of a sample lies along some fixed direction(s). Strong easy-axis anisotropy is a prerequisite for hard magnetism. Near-zero anisotropy is required for soft magnets. Generally, the tendency for magnetization to lie along an easy axis is represented by the energy density term introduced in Chapter 1:

$$E_a = K_1 \sin^2 \theta, \quad (5.62)$$

where θ is the angle between $\mathbf{M}$ and the anisotropy axis. K_1 has units J m^{-3} . Values may range from less than 1 kJ m^{-3} to more than 20 MJ m^{-3} . The anisotropy depends on temperature, and must tend to zero at T_C if there is no applied field. Three main sources of anisotropy are related to sample shape, crystal structure and atomic or micro-scale texture.

Shape anisotropy derives from the demagnetizing field discussed in §2.2.4. The energy of a sample in its demagnetizing field H_d gives a contribution to the self-energy (§2.5.1), which depends on the direction of magnetization in the sample. Obviously, this cannot be an intrinsic property of the material, as it depends on the sample shape.

Magnetocrystalline anisotropy is an intrinsic property. The magnetization process is different when the field is applied along different crystallographic directions, and the anisotropy reflects the crystal symmetry. Its origin is in the crystal-field interaction and spin-orbit coupling, or else the interatomic dipole-dipole interaction.

Induced anisotropy arises when an easy direction of magnetization is created by applied stress, or by depositing or annealing a disordered alloy in a magnetic field to create some atomic-scale texture. The texture is often subtle, and difficult to discern by conventional X-ray or electron scattering methods. Coarser-scale texture may be associated with mesoscopic fluctuations of the compositions, such as spinodal decomposition.

We first consider the phenomenological expressions for E_a , and then look deeper into its microscopic origins.

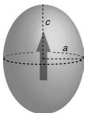
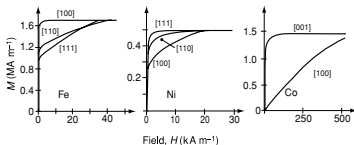
5.5.1 Shape anisotropy

The magnetostatic energy of a ferromagnetic ellipsoid with magnetization M_s is

$$\varepsilon_m = \frac{1}{2} \mu_0 V \mathcal{N} M_s^2. \quad (2.81)$$

Figure 5.27

Magnetization of single crystals of iron, cobalt and nickel.



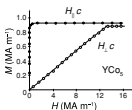
Magnetization of a prolate ellipsoid of revolution with $c > a$ and no magnetocrystalline anisotropy. The c axis is the easy direction of magnetization.

Other simple shapes can be approximated to ellipsoids. The anisotropy energy is related to the difference in energy $\Delta\epsilon$ when the ellipsoid is magnetized along its hard and easy directions. $\mathcal{N}$ is the demagnetizing factor for the easy direction; $\mathcal{N}' = \frac{1}{2}(1 - \mathcal{N})$ is the demagnetizing factor for the perpendicular, hard, directions (Table 2.1). Hence $\Delta\epsilon_m = \frac{1}{2}\mu_0 V M_s^2 [\frac{1}{2}(1 - \mathcal{N}) - \mathcal{N}']$, which gives for a prolate ellipsoid

$$K_{sh} = \frac{1}{4}\mu_0 M_s^2 (1 - 3\mathcal{N}). \quad (5.63)$$

This is zero for a sphere ($\mathcal{N} = \frac{1}{3}$), as expected. Shape anisotropy is only fully effective in samples which are so small that they do not break up into domains. Non-ellipsoidal shapes are approximately described by an effective demagnetizing factor. In a multidomain state, each domain creates its own demagnetizing field, and is subject to the stray field of the other domains. The order of magnitude of shape anisotropy for a ferromagnet with $\mu_0 M_s \approx 1$ T is 200 kJ m^{-3} when $\mathcal{N} = 0$.

5.5.2 Magnetocrystalline anisotropy



Magnetization of a crystal of the intermetallic compound YCo_5 . K_1 is deduced from hard axis saturation.

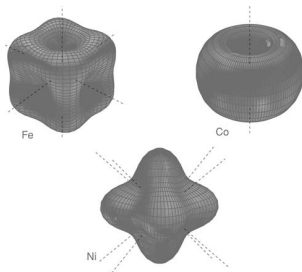
The magnetization curves of single crystals of the three 3d ferromagnetic elements, corrected for the demagnetizing field, show a different approach to saturation when magnetized in different directions (Fig. 5.27). For iron, the cube edges $\{100\}$ are easy directions and the cube diagonals $\{111\}$ are hard directions. For nickel it is the other way around. Cobalt, and many intermetallic compounds such as YCo_5 have the hexagonal axis $[001]$ as the unique easy direction. It is much more difficult to saturate YCo_5 , for example, than Co in a hard direction, perpendicular to $[001]$. Uniaxial anisotropy is a prerequisite for permanent magnetism.

The three-dimensional anisotropy surfaces for the three ferromagnetic elements are shown in Fig. 5.28.

³ $[\]$ denotes a single direction and $\{ \}$ denotes a set of equivalent directions. Similarly $()$ denotes a plane and $\{ \}$ a set of equivalent planes.

Figure 5.28

Magnetocrystalline anisotropy energy surfaces for iron, cobalt and nickel. In iron there are three easy axes (100), in cobalt one [001] and in nickel four (111).



Conventional expressions for the anisotropy energy in different symmetries are:

$$\text{Hexagonal: } E_a = K_1 \sin^2 \theta + K_2 \sin^4 \theta + K_3 \sin^6 \theta + K'_3 \sin^6 \theta \sin 6\phi,$$

$$\text{Tetragonal: } E_a = K_1 \sin^2 \theta + K_2 \sin^4 \theta + K'_2 \sin^4 \theta \cos 4\phi + K_3 \sin^6 \theta + K'_3 \sin^6 \theta \sin 4\phi,$$

$$\text{Cubic: } E_a = K_{1c}(\alpha_1^2 \alpha_2^2 + \alpha_2^2 \alpha_3^2 + \alpha_3^2 \alpha_1^2) + K_{2c}(\alpha_1^2 \alpha_2^2 \alpha_3^2),$$

where α_i are the direction cosines of the magnetization.

The K_{1c} term is equivalent to $K_{1c}(\sin^4 \theta \cos^2 \phi \sin^2 \phi + \cos^2 \theta \sin^2 \phi)$. When $\theta \approx 0$, $\phi = 0$ this reduces to $K_{1c} \sin^2 \theta$, so the leading term in every case is of the form (5.62).

An alternative way of writing the anisotropy expressions is in terms of a set of orthonormal spherical harmonics with anisotropy coefficients κ_l^m , and the crystal-field coefficients A_l^m introduced in §4.4:

$$E_a = \sum_{l=2,4,6} \kappa_l^m A_l^m Y_l^m(\theta, \phi). \quad (5.64)$$

For example,

$$\text{Hexagonal: } E_a^{\text{hex}} = \kappa_0 + \kappa_2^0(\alpha^2 - \frac{1}{3}) + \kappa_4^0(\alpha^4 - \frac{6}{7}\alpha^2 + \frac{3}{35}) + \dots$$

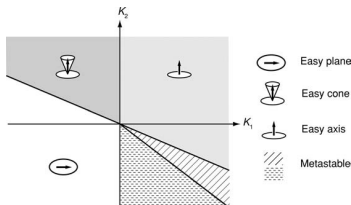
with $\alpha = \cos \theta$.

$$\text{Cubic: } E_a^{\text{cubic}} = \kappa_0 + \kappa_4^4(\alpha_1^2 \alpha_2^2 + \alpha_2^2 \alpha_3^2 + \alpha_3^2 \alpha_1^2 - \frac{1}{5}) + \kappa_6^4(\alpha_1^4 \alpha_2^2 \alpha_3^2 - \frac{1}{11}(\alpha_1^2 \alpha_2^2 + \alpha_2^2 \alpha_3^2 + \alpha_3^2 \alpha_1^2 - \frac{1}{5}) - \frac{1}{105})$$

This shows clearly that K_{1c} actually relates to fourth-order anisotropy terms, whereas K_1 in uniaxial structures relates to second-order anisotropy.

Figure 5.29

Magnetic phase diagram for uniaxial magnets. In the metastable regions, there are two energy minima, at $\theta = 0$ and $\theta = \pi/2$.



An interesting magnetic phase diagram (Fig. 5.29) arises for a uniaxial magnet when both K_1 and K_2 are taken into account; minimizing the anisotropy energy $E_a = K_1 \sin^2 \theta + K_2 \sin^4 \theta$, an easy cone phase appears when $K_1 < 0$ and $K_2 > -K_1/2$. The cone angle is $\sin^{-1} \sqrt{(|K_1|/2K_2)}$. In practice, when $K_1 > 0$, the two anisotropy constants K_1 and K_2 can be deduced from a plot of the hard axis magnetization curve as H/M versus M^2 . This is the Sucksmith–Thomson plot.

The anisotropy field H_a is defined as the field needed to saturate the magnetization of a uniaxial crystal in a hard direction:

$$E = K_u \sin^2 \theta - \mu_0 M_s H \cos(\pi/2 - \theta).$$

Minimizing E , $\partial E/\partial \theta = 0$ and setting $\theta = \pi/2$,

$$H_a = 2K_u/\mu_0 M_s. \quad (5.65)$$

Since $\mu_0 M_s \approx 1$ T for a typical ferromagnet, H_a can range from <2 kA m $^{-1}$ to >20 MA m $^{-1}$, with typical values for shape anisotropy for $\mathcal{N} = 1$ of 200 kA m $^{-1}$. An equivalent definition of H_a is the applied field along the easy axis which would reproduce the change in energy due to a small deviation of the magnetization from this axis. This gives $K_u \sin^2 \delta \theta = \mu_0 H_a M_s (1 - \cos \delta \theta)$, which returns the same result as (5.65).

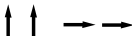
Beware of taking the idea of anisotropy field too literally. Except at small angles, the energy variation in a field is not the same as the leading term in the anisotropy. A magnetic field defines an easy direction, not an easy axis.

5.5.3 Origin of magnetocrystalline anisotropy

There are two distinct sources of magnetocrystalline anisotropy:

- single-ion contributions;
- two-ion contributions.

Single-ion anisotropy The single-ion contribution is essentially due to the electrostatic interaction of the orbitals containing the magnetic electrons with the potential created at the atomic site by the rest of the crystal. This crystal-field interaction tends to stabilize a particular orbital, and by spin-orbit interaction the magnetic moment is aligned in a particular crystallographic direction. The single-ion contribution was discussed in §4.4, where its effect on the paramagnetic susceptibility was described. In a ferromagnetic crystal, the contributions of all the ions are summed to produce a set of macroscopic energy terms with appropriate symmetry. The sum is straightforward when the local anisotropy axes of all the sites in the unit cell coincide. For example, a uniaxial crystal having $n = 2 \times 10^{28}$ ions m^{-3} , described by a spin Hamiltonian $D\hat{S}_z^2$ with $D/k_B = 1$ K and $S = 2$ will have anisotropy constant $K_1 = nDS^2 = 1.1 \times 10^6$ J m^{-3} .



Broadside and head-to-tail configurations for a pair of ferromagnetically coupled magnetic moments. The latter is lower in energy.

Two-ion anisotropy The two-ion contribution often reflects the anisotropy of the dipole-dipole interaction. Comparing the broadside and head-to-tail configurations of two dipoles, each with moment m , it can be seen from (2.10) and (2.73) that the head-to-tail configuration is lower in energy by $3\mu_0 m^2 / 4\pi r^3$. Magnets tend to align head-to-tail. This anisotropy is of order 1 K per atom, or 100 kJ m^{-3} . However, the dipole sum has to be extended over the entire lattice, and it vanishes for certain lattices (including all the cubic lattices). In noncubic lattices, the dipole interaction is an appreciable source of ferromagnetic anisotropy.

Another source of two-ion anisotropy is anisotropic exchange. The Heisenberg Hamiltonian is perfectly isotropic, but there are higher-order corrections involving spin-orbit coupling that lead to preferred orientations of the exchange-coupled pairs.

5.5.4 Induced anisotropy

One way of inducing uniaxial anisotropy is to anneal certain alloys in a magnetic field, Fig. 5.30. A good example is permalloy, $\text{Ni}_{80}\text{Fe}_{20}$, which is an fcc alloy with the property that $K_1 \approx 0$. When annealed at ~ 800 K, atomic diffusion proceeds in the sense of favouring head-to-tail pairs of iron atoms, which have a larger moment than nickel. The iron pairs tend to align with the field, Fig. 5.31. This built-in texture produces a weak anisotropy. Amorphous ferromagnets may likewise acquire a uniaxial anisotropy due to pairwise texture when annealed in a magnetic field. Similar texture can be achieved by atomic deposition of thin films in a magnetic field.

Another way to induce uniaxial anisotropy in a ferromagnetic solid is to apply uniaxial stress σ . The magnitude of the stress-induced anisotropy is $K_{H\sigma} = \frac{3}{2}\sigma\lambda_s$, where λ_s is the saturation magnetostriction discussed in the next

Figure 5.30

Magnetization of a thin film with induced anisotropy created by annealing in a magnetic field. The 'sheared' loop on the left is found when the measuring field H' is applied perpendicular to the annealing field direction. The open loop on the right is obtained when the two directions are parallel.

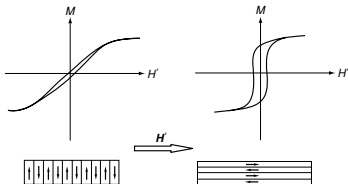
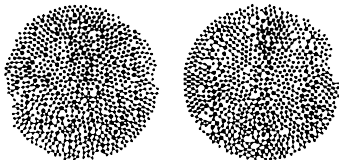


Figure 5.31

Pairwise texture induced by magnetic annealing. Pairs of atoms represented by the larger dots tend to be aligned vertically (G. S. Cargill and T. Mizoguchi, *J. Appl. Phys.* 49 1753 (1978)).



section. Both the single-ion and two-ion anisotropy contribute to the stress-induced effect.

The largest values of uniaxial anisotropy are found in hexagonal and other uniaxial crystals. Smallest values are found in certain cubic alloys and amorphous ferromagnets. The magnitudes of different anisotropy contributions are summarized in Fig. 5.32.

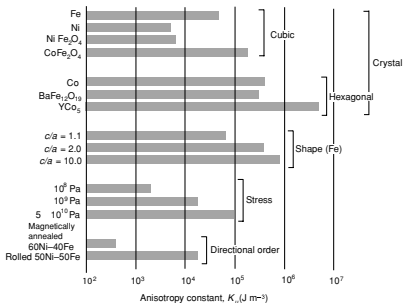
5.5.5 Temperature dependence

Any anisotropy originating from the magnetic dipole interaction will vary with temperature as M^2 because the dipole field $H_d \propto M_s$ and the energy density $E_a = -\frac{1}{2}\mu_0 \mathbf{M}_s \cdot \mathbf{H}_d \propto M_s^2$.

Single-ion anisotropy due to the crystal field exhibits a $j(j+1)/2$ power law at low temperatures for the second-, fourth- and sixth-order terms, which vary respectively as the third, tenth and twenty-first powers of the magnetization; $K_j(T)/K_j(0) = M^{j(j+1)/2}$. Close to T_C , these turn into M^j power laws. Here is a rare example of an effect whose temperature dependence is determined by symmetry alone.

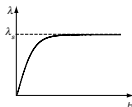
Figure 5.32

Quantitative overview of magnetic anisotropy, from all sources. (After Cullity and Graham (2008))



5.6 Ferromagnetic phenomena

Magnetic order in a crystal can influence the lattice parameters and the elastic moduli. These effects reflect the volume or strain dependence of bandwidth, exchange energy, magnetic dipole energy or crystal-field interaction. Some effects, like linear magnetostriction, are small, $\lambda_s \approx 10^{-5}$. Others like the ΔE effect, the change of Young's modulus associated with domain alignment, can be as large as 90%. Even very small magnetoelastic effects may be critical when optimizing the properties of soft magnets, because of the sensitivity of coercivity to stress-induced anisotropy. Ferromagnetism also influences the thermal, electrical and optical properties of solids. These *intrinsic* effects in ferromagnets are discussed in the following paragraphs.



Development of magnetostriction of a polycrystalline material as a function of applied magnetic field.

5.6.1 Magnetostriction

Spontaneous volume magnetostriction ω_s is the fractional change of volume of an isotropic crystal due to magnetic order; the effect is proportional to M^2 . The sign may be positive or negative but the magnitude does not normally exceed 1%. Iron-rich fcc alloys, for example, have $\omega_s > 0$, reflecting the sensitivity of exchange to interatomic spacing (Fig. 5.15). By expanding slightly, the iron-rich alloy increases the ferromagnetic exchange interaction, and lowers its energy.

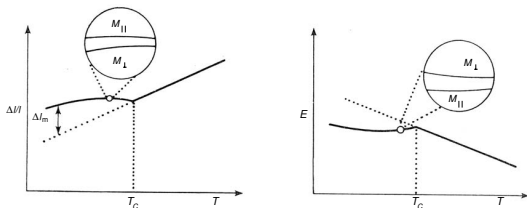
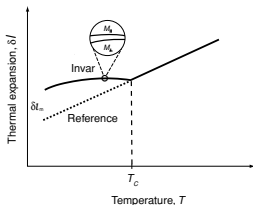


Figure 5.33

The invar effect; magnetostriction, Young's modulus and thermal expansion of an iron-based ferromagnetic alloy. The dotted line shows the thermal expansion of a reference material, while the thermal expansion of an invar alloy is shown by the solid line. The difference is the magnetovolume anomaly.



Invar is an fcc iron–nickel alloy of composition $\text{Fe}_{65}\text{Ni}_{35}$, discovered by Charles Guillaume in 1896. The alloy has zero thermal expansion because of a positive magnetovolume anomaly. Its discovery was important at the time because of the practical uses for alloys that are dimensionally stable in precision instruments. By varying the Fe:Ni ratio slightly, a match can be achieved with the thermal expansion of other materials, such as Si or quartz. The spontaneous volume magnetostriction of a ferromagnet is inferred by comparison with the thermal expansion of a nonmagnetic reference material, as shown in Fig. 5.33.

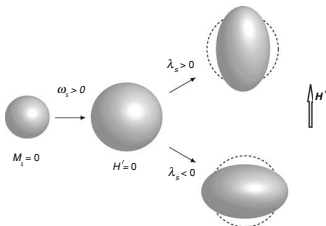
In weak ferromagnets, there is also a forced volume magnetostriction in high magnetic fields, as the moment is slightly increased by additional field-induced band splitting.

Table 5.7. Summary of magnetoelastic effects

	Isotropic	Anisotropic
Atomic spacing	Spontaneous volume magnetostriction Forced volume magnetostriction Anomalous thermal expansion	Spontaneous linear magnetostriction
Elastic moduli	Anomalous elastic moduli	Morphic effect, ΔE effect

Figure 5.34

Spontaneous volume and linear magnetostriction.



James Joule 1818–1889.

Linear magnetostriction was discovered by James Joule in nickel in 1842. The linear strain $\delta l/l$ in the direction of magnetization is associated with the magnetization process. Saturation magnetostriction is denoted as λ_s . Iron has $\lambda_s = -7 \times 10^{-6}$ (the dimensionless unit of 10^{-6} is known as a *microstrain*), so it contracts by 8 ppm along the magnetic axis as the magnetization is saturated. Volume is conserved, which means that it also expands by 4 ppm in the perpendicular direction and $\lambda_{||} + 2\lambda_{\perp} = 0$. Thanks to magnetostriction, no ferromagnetically ordered crystal can ever be strictly cubic! Linear magnetostriction is the reason why transformers hum.

In general, if θ is the angle between the magnetization and the easy axis,

$$\lambda(\theta) = \lambda_s(3 \cos^2 \theta - 1)/2.$$

The main magnetostrictive effects, summarized in Table 5.7, are greatly exaggerated in Fig. 5.34.

Linear magnetostriction depends on the direction of magnetization relative to the crystal axes, although, not of course, on the sense of its orientation along a particular direction. A change of strain is associated with rotation of domains as the magnetization approaches saturation, not with motion of 180° domain

Table 5.8. Summary of linear magnetostrictive effects

Joule effect	Villari effect
Field-induced strain	Strain-induced anisotropy
Wiedemann effect	Matteuchi effect
Helical-field-induced torque	Torque-induced anisotropy

walls. Magnetization reversal within a domain produces no strain. A single parameter is sufficient to describe the linear magnetostriction of an isotropic polycrystalline or amorphous material, but magnetostriction is really a third-rank tensor, since it relates strain ϵ_{ij} (a second-rank tensor) and magnetization M_k (a vector – first-rank tensor). In cubic crystals, there are two principal values. For example, in iron $\lambda_{100} = 15 \times 10^{-6}$ and $\lambda_{111} = -21 \times 10^{-6}$. The isotropic average,

$$\lambda_s = \frac{2}{3}\lambda_{100} + \frac{1}{3}\lambda_{111},$$

is 7×10^{-6} . Larger magnetostriction is found in nickel, where Joule first saw the effect, and in cobalt (Table 5.4). Some mixed rare-earth–iron alloys exhibit λ_s values of up to 2000×10^{-6} . The converse effect, where mechanical stress applied to a demagnetized multidomain ferromagnet modifies its easy axes and changes the initial susceptibility is known as the Villari effect.

A variant of linear magnetostriction is the Wiedemann effect. A ferromagnetic rod subject to a transverse helicoidal field tends to twist. Torque can therefore be generated by the field produced by passing a current through a ferromagnetic wire while applying a field along its axis. The twist angle ϕ is proportional to the length of the wire l , the current density j and the magnetostriction λ_s :

$$\phi = \frac{3}{2}\lambda_s j l H_{\parallel}. \quad (5.66)$$

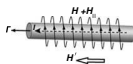
The effect can be used to measure magnetostriction.

The converse of the Wiedemann effect, known as the Matteuchi effect, is the modification of the susceptibility of a ferromagnetic wire by torque. The Joule and Wiedemann effects find applications in various types of magnetic actuators, whereas the Villari and Matteuchi effects are exploited in electromechanical sensors. The linear magnetostrictive effects are summarized in Table 5.8.

5.6.2 Other magnetoelastic effects

The effect of imposing a uniaxial stress $\sigma \text{ N m}^{-2}$ on a ferromagnetic material is to create strain-induced anisotropy. The elastic energy density for a polycrystalline material is

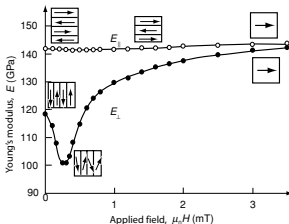
$$E_{ms} = -\lambda_s(E/2)(3 \cos^2 \theta - 1)\epsilon + (1/2)E\epsilon^2,$$



The Wiedemann effect. The sense of the twist shown is for $\lambda_s > 0$.

Figure 5.35

The ΔE effect in a soft, magnetostrictive ferromagnet. E is Young's modulus. The inserts show the evolution of the domain structure when stress is exerted in an applied field. (after B. S. Berry and W. C. Pritchett, *Phys Rev. Lett* **34**, 1022 (1975))



where ϵ is the strain, E is Young's modulus and θ is the angle between the magnetization and the strain axis. Minimizing the energy with respect to ϵ gives $\epsilon = \lambda_s / 2(3 \cos^2 \theta - 1)$. If no stress is applied, the equilibrium strain $\epsilon = \lambda_s$. Since $\sigma = \epsilon E$, the angular-dependent term is

$$E_{ms} = -\lambda_s (\sigma / 2) (3 \cos^2 \theta - 1). \quad (5.67)$$

Comparing with the usual expression for uniaxial anisotropy energy $E_a = K_u \sin^2 \theta$ (5.62), we see that

$$K_u = \frac{3}{2} \lambda_s \sigma, \quad (5.68)$$

as mentioned in §5.5.4. In iron $\lambda_s \approx -7 \times 10^{-6}$ and $E \approx 200$ GPa, hence a stress of 1.4 MPa (14 bars) can be created by the magnetostriction.

The coupling of strain and magnetostriction is manifest in a dramatic way in the ΔE effect. The magnetization directions of domains in an unsaturated ferromagnet will be modified by an applied stress, which in turn leads to a magnetostrictive strain in the direction of the stress. This looks like a change in Young's modulus E , hence the name. Regardless of the sign of λ , the effect is always to reduce E , which is defined as the ratio of linear stress to strain (σ / ϵ). If a tensile stress is applied to a material with positive λ , the magnetization directions align with the stress, and there is an elongation, which increases ϵ and reduces E . If the tensile stress is applied to a material with negative λ , the magnetization directions align perpendicular to the stress, producing an elongation which is half as large, but again reduces E . The effect disappears as soon as the magnetization is saturated along or perpendicular to the stress direction. Some applications of the ΔE effect (Fig. 5.35) in ferromagnetic thin films are presented in Chapter 12.

The morphic effect is the dependence of the elastic constants on the direction of magnetization of a ferromagnet.

Table 5.9. Summary of uniaxial anisotropy

Anisotropy	Energy	K_u	kJ m^{-3}
Magnetocrystalline	Crystal field	K_1	$1-10^4$
	Magnetic dipole		
Shape	Magnetostatic (magnetic dipole)	$\frac{1}{2}\mu_0 M_s^2 [3N-1]$	$1-500$
Stress	Magnetoelastic	$3\lambda_s \sigma / 2$	$0-100$

Table 5.10. Energy contributions in a ferromagnet, in kJ m^{-3}

Exchange	$-2n\mathcal{J}ZS^2$	10^3-10^5
Anisotropy	K_u, K_i	$10^{-1}-10^4$
Magnetic self-energy	E_d	$0-2 \times 10^3$
External field energy (in 1 T)	$B_0 M_s$	10^2-10^3
External stress energy (in 1 GPa)	$\sigma \lambda$	$1-10^2$
Magnetostrictive self-energy	$c\lambda^2$	$0-1$

10^3 kJ m^{-3} is roughly equivalent to 1 K or 0.1 meV atom⁻¹

The sources of uniaxial anisotropy are summarized in Table 5.9. A summary of the magnitudes of the energies involved in ferromagnetism is provided in Table 5.10.

5.6.3 Magnetocaloric effect

We have seen that a specific heat of magnetic origin is associated with the progressive disordering of atomic moments in a magnetically ordered material on increasing the temperature. There is a λ anomaly at the Curie point. The contributions to the total specific heat of nickel are illustrated in Fig. 5.36. The exchange energy, E_{ex} in the molecular field model is $-\frac{1}{2}\mu_0 H^i M_s$, where $H^i = n_W M_s$. Hence the specific heat of magnetic origin, $C_m = dE_{ex}/dT = (dE_{ex}/dM_s)(dM_s/dT)$, giving

$$C_m = -\mu_0 n_W M_s \frac{dM_s}{dT}. \quad (5.69)$$

In this model, there is no magnetic contribution above T_C where $M_s = 0$. However, short-range spin correlations persist above T_C . A more accurate expression for the specific heat, deduced from (5.24) for nearest-neighbour exchange coupling in terms of the spin correlation function $\langle \mathbf{S}_i \cdot \mathbf{S}_j \rangle$, is

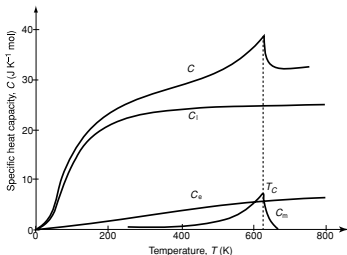
$$C_m = -2nZ\mathcal{J} \frac{\partial \langle \mathbf{S}_i \cdot \mathbf{S}_j \rangle}{\partial T}.$$

where n is the number of atoms per unit volume.

The persistence of magnetic specific heat above T_C is evident in Fig. 5.36.

Figure 5.36

Heat capacity of nickel, showing electronic, lattice and magnetic contributions, and their sum C .



The effect of an applied magnetic field is to reduce magnetic disorder, and decrease magnetic entropy. If this is done adiabatically, with no exchange of heat with the surroundings, there has to be an increase of temperature of the sample. Conversely, if the thermally isolated sample is magnetized in an applied field, which is then reduced to zero, there must be a decrease of temperature in the sample due to adiabatic demagnetization, as described for paramagnets in §4.3.4. The entropy change is related to the temperature derivative of the Magnetization by Eq. 2.102.

The work done by the field to produce an increase of magnetization δM is $\delta W = \mu_0 H' \delta M$ (2.93), so the heat resulting from the change of magnetization, which is the difference between this work and the change of exchange energy $-\mu_0 n_W M \delta M$, is

$$\delta Q = \mu_0 (H' + n_W M) \delta M. \quad (5.70)$$

In the paramagnetic temperature range above T_C , $\chi = C/(T - T_C)$, which from (5.6) is $T_C/[n_W(T - T_C)]$, hence

$$M = \frac{H' T_C}{n_W (T - T_C)}. \quad (5.71)$$

Substituting (5.71) into (5.70), it follows that

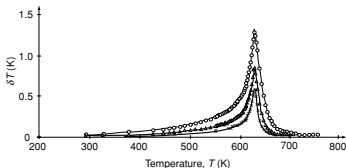
$$\delta Q = \frac{\mu_0}{2n_W} \frac{T T_C}{(T - T_C)^2} \delta(H^2).$$

The corresponding change of temperature is $\delta T = \delta Q/C_M$, where C_M is the specific heat at constant magnetization (in units of $\text{J K}^{-1} \text{m}^{-3}$). Expressed in terms of the change of magnetization, we find from (5.71) for $T > T_C$:

$$\delta T = \frac{\mu_0 n_W}{2C_M} \frac{T}{T_C} \delta(M^2).$$

Figure 5.37

The magnetocaloric effect for Ni in different applied fields. It peaks at T_C .



The result for $T < T_C$ is almost the same; it is obtained by neglecting the applied field H' relative to the molecular field $n_W M$ in the expression for δQ

$$\delta T = \frac{\mu_0 n_W}{2C_M} \delta(M^2).$$

The magnetocaloric effect for Ni in a field of 2 T is illustrated in Fig. 5.37. The temperature change at T_C is of order 2 K in a field of 1 T. Gd-based alloys with T_C as 320 K may be used for magnetic refrigeration.

5.6.4 Magnetotransport

Here we consider *intrinsic* magnetotransport effects which do not depend on the shape or form of a ferromagnetic sample. Several types of magnetoresistive and Hall effects fall into this category. Magnetoresistance may be defined as

$$MR = [\varrho(B) - \varrho(0)]/\varrho(0). \quad (5.72)$$

The ordinary positive B^2 magnetoresistance of a nonmagnetic metal due to cyclotron motion of the electrons was mentioned in §3.2.7.

There are other magnetoresistance phenomena intrinsic to ferromagnets, notably anisotropic magnetoresistance (AMR), a small effect depending on the relative orientation of current and magnetization, and colossal magnetoresistance (CMR) a much larger negative effect which appears in the vicinity of the Curie point of materials where the exchange coupling is by double exchange, and the electron transfer is linked to nearest-neighbour spin alignment. A further source of negative magnetoresistance in ferromagnets is spin disorder scattering in the vicinity of T_C . Ferromagnets also exhibit an additional term in the Hall effect which is proportional to the magnetization.

Conduction electrons in 3d ferromagnets and their alloys are in *s*-like or *d*-like states, which coexist at the Fermi level. The *s*-electrons resemble free electrons; they have high mobility and carry most of the current. The *d*-electrons are relatively ineffective current carriers on account of their high effective mass

m^* and low mobility, $\mu = e\tau/m^*$, where τ is the scattering relaxation time. The d -bands are spin-split, with different densities of states at the Fermi level for the $\uparrow$ and $\downarrow$ subbands. The s -bands are not spin-split, but s states at the Fermi level acquire predominantly $\uparrow$ or $\downarrow$ character by hybridization with d states there. Scattering is more severe for the $\downarrow$ s -electrons in a strong ferromagnet, because they can only be scattered into $3d^\downarrow$ states if spin-flip scattering is neglected. In a weak ferromagnet electrons of either spin can be scattered into unoccupied d states of the same spin. The densities of state of copper and nickel were compared in Fig. 5.15. The conduction electrons in copper are s -like and weakly scattered, leading to low resistivity. In nickel, a strong ferromagnet, the $\uparrow$ states at the Fermi level conduct like copper, but the $\downarrow$ states are very strongly scattered on account of the large density of d states at the Fermi level into which they can scatter.

The situation in $4f$ metals and alloys is different. The $4f$ -band is usually very narrow because of the limited overlap of $4f$ wave functions on adjacent sites. There is integral $4f^n$ electron occupancy, so the band does not conduct. The Mott criterion for conduction in a narrow band is $W > U$, where W is the bandwidth and U is the screened Coulomb energy of an extra electron on the atomic site. When $W < U$, the charged configurations $4f^{n\pm 1}$ necessary for metallic conduction cannot be accommodated in the band, and the material is a Mott-Hubbard insulator. Conduction in the rare-earth metals and alloys is therefore due to the partially occupied $5d$ - and $6s$ -orbitals at the Fermi level. The $5d$ states are exchange-split by on-site $4f$ - $5d$ exchange coupling. The electronic configuration of a rare-earth is roughly $4f^n 5d^2 6s$, so they behave in some ways like light d -elements.

The $5f$ -band of ferromagnetic compounds of the early actinides is broad and conducting, as in the $3d$ metals, but for the heavy actinides there is integral $5f$ occupancy, and the f electrons are localized, as in the rare-earths.

For ionic compounds, such as oxides, the $3d$ band is narrow and the electrons tend to be localized. There are a few exceptions such as CrO_2 or SrRuO_3 where the band is broad enough for $3d^n$ and $3d^{n\pm 1}$ configurations to coexist. Thanks to the on-site exchange interaction, these d states are spin polarized and $3d^{n+1} \rightleftharpoons 3d^n$ electron hopping produces ferromagnetism, via the double exchange interaction. Mobility is low, and the velocity of the electrons may be slow enough in oxides for the Born-Oppenheimer approximation to be invalid. The electrons drag a local lattice distortion along with them, forming a polaron, which further enhances their effective mass. When the conduction electrons are spin polarized, they drag a cloud of spin polarization of the $3d$ or $4f$ ion cores along with them; then we have a spin polaron.

The $\uparrow$ or $\downarrow$ electrons in a ferromagnet may undergo spin-flip scattering events, associated with excitation or absorption of a magnon ($\Delta S = 1$). The spin-orbit interaction also mixes $\uparrow$ and $\downarrow$ channels, because $\mathbf{L} \cdot \mathbf{S} = L_x S_x + L_y S_y + L_z S_z$ can be written in terms of raising and lowering operators as $\frac{1}{2}(L^+ S^- + L^- S^+) + L_z S_z$. It therefore mixes $|\frac{1}{2}, m_\ell\rangle$ with $|\frac{1}{2}, m_\ell + 1\rangle$

Table 5.11. Room-temperature resistivity of metals ($10^{-8} \Omega \text{ m}$)

Metal	Orbitals	Magnetization	$\varrho_{\uparrow}$	$\varrho_{\downarrow}$	ϱ	α
Cu	<i>s</i> -band	Paramagnet	4	4	2	1
Ni	<i>d</i> -band	Strong ferromagnet	13	65	11	5
Co	<i>d</i> -band	Strong ferromagnet	8	120	7	15
Fe	<i>d</i> -band	Weak ferromagnet	32	28	15	0.9
V	<i>d</i> -band	Paramagnet	52	52	26	1
aFe ₈₀ B ₂₀	<i>d</i> -band	Amorphous ferromagnet	320	320	120	1
Gd	<i>f</i> -level; 5 <i>d</i> /5 <i>s</i> -band	Ferromagnet	270	270	130	1

states. Nevertheless, these spin flipping events are relatively rare compared with normal momentum scattering events. An electron in a ferromagnetic 3*d* metal may undergo 100 or more scattering events, before it experiences a spin flip. This led Mott to propose his two-current model in 1936, whereby the $\uparrow$ and $\downarrow$ conduction channels are regarded as independent, and conduct in parallel. Hence $\sigma = \sigma_{\uparrow} + \sigma_{\downarrow}$, or in terms of resistivity

$$\varrho = \frac{\varrho_{\uparrow}\varrho_{\downarrow}}{(\varrho_{\uparrow} + \varrho_{\downarrow})}. \quad (5.73)$$

The quantity α is defined as the ratio of conductivity in the $\uparrow$ and $\downarrow$ channels: $\alpha = \sigma_{\uparrow}/\sigma_{\downarrow} = \varrho_{\downarrow}/\varrho_{\uparrow}$. The conductivity of a metal is always greater than the conductivity of either spin channel – double when the metal is paramagnetic and $\sigma_{\uparrow} = \sigma_{\downarrow}$. Values of $\varrho_{\uparrow,\downarrow}$ and α for a few metals are given in Table 5.11.

A useful result for the field dependence of the resistivity of a metal where the resistivity can be described by a single scattering time is Kohler's rule:

$$\frac{\varrho(H)}{\varrho(0)} = f\left(\frac{H}{\varrho(0)}\right)^2. \quad (5.74)$$

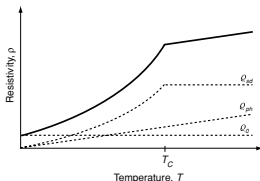
It follows that $\Delta\varrho/\varrho \approx (H/\varrho)^2$.

Spin-disorder scattering Electrons in ferromagnetic solids experience a spin-dependent potential. When the material is perfectly ordered, there is no magnetic scattering. But as the temperature approaches T_C the electrons see a potential that fluctuates between values differing by the on-site exchange interaction. This random potential contributes to the resistivity as shown in Fig. 5.38.

The normal temperature variation of the resistivity of a metal is described by Matthiessen's rule, a statement that the resistivity is the sum of a temperature-independent term ϱ_0 due to impurity scattering, which is of order 10 n Ω m per % of impurity, and a temperature-dependent term $\varrho_{ph}(T)$ due to phonon scattering, which varies as T^n , where $n = 1$ at temperatures well above the

Figure 5.38

The resistivity of a normal metal showing the impurity contribution ϱ_0 and the phonon contribution ϱ_{ph} . An extra term due to spin-disorder scattering ϱ_{sd} appears in a ferromagnet.



Debye temperature and $n = 3-5$ far below it. In ferromagnets, there is an extra impurity-like term due to disordered spins. When majority-spin electrons, which carry most of the current, encounter an atom with reduced spin moment, they are strongly scattered because of the increased local exchange potential. The spin disorder scattering term in the paramagnetic state varies as

$$\varrho_{para} \sim \frac{k_F m_e^2 \mathcal{J}_{sd}^2 s(s+1)}{e^2 \hbar^2}, \quad (5.75)$$

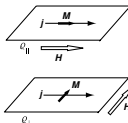
where $\mathcal{J}_{sd}$ is the exchange between localized and conduction electrons. This leads to an expression for the spin disorder resistivity in the ferromagnetic state

$$\varrho_{ferro} = \varrho_{para} \{1 - [M_s(T)/M_s(0)]^2\}. \quad (5.76)$$

Spin-disorder scattering is pronounced in rare-earth metals and alloys, and in half-metals. Applying a magnetic field close to T_C where the susceptibility is high, increases M , and therefore produces a negative magnetoresistance.

An alternative way to describe spin mixing is to introduce a spin mixing resistivity $\varrho_{\uparrow\downarrow}$ which is zero at low temperature, but large above T_C . The expression for resistivity is then

$$\varrho = \frac{\varrho_{\uparrow}\varrho_{\downarrow} + \varrho_{\uparrow\downarrow}(\varrho_{\uparrow} + \varrho_{\downarrow})}{\varrho_{\uparrow} + \varrho_{\downarrow} + 4\varrho_{\uparrow\downarrow}}. \quad (5.77)$$



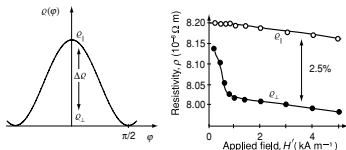
Measurement of AMR for a thin film.

This has the correct behaviour in the high-temperature limit, where it tends to $(\varrho_{\uparrow} + \varrho_{\downarrow})/4 = \varrho_{para}$, and in the low-temperature limit where it reduces to (5.73).

Anisotropic magnetoresistance AMR is a much smaller change, but it is useful for sensors because the resistance change can be achieved in thin films with magnetic fields of a millitesla or less. William Thompson discovered in 1857 that the resistivity of nickel varies slightly with the direction of current flow, relative to the direction of magnetization. The resistance is a few per cent higher when the current flows parallel to an applied field than when it is in

Figure 5.39

AMR of a permalloy film, and its angular dependence.



the transverse direction. Anisotropy in this sense is usual for ferromagnetic metals. The maximum AMR may be defined as $\Delta\rho/\rho_{\perp}$, where $\Delta\rho = (\rho_{\parallel} - \rho_{\perp})$. The magnitude of the effect is usually no more than about 3%. The angular dependence of the saturated anisotropic magnetoresistance is

$$\rho(\varphi) = \rho_{\perp} + (\rho_{\parallel} - \rho_{\perp}) \cos^2 \varphi, \quad (5.78)$$

where φ is the angle between the current density $\mathbf{j}$ and the magnetization $\mathbf{M}_s$. Results for permalloy are shown in Fig. 5.39.

Note that the greatest sensitivity of the resistance to a change in φ , determined by the condition $d^2\rho/d^2\varphi = 0$, arises when $\varphi = \pi/4$, a fact that is exploited in AMR sensors, discussed in §14.3.

When the current flows in the x -direction, an easy-anisotropy direction – this could be determined by the shape anisotropy of a wire, for example – with anisotropy constant K_u and the field applied in the perpendicular y -direction, then φ is obtained by minimizing

$$E = M_s H' \sin \varphi + K_u \cos^2 \varphi \quad (5.79)$$

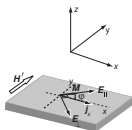
with respect to φ , which gives $\sin \varphi = M_s H' / 2K_u$. Hence, from (5.78)

$$\rho(H) = \rho_{\perp} \left[1 + \frac{\Delta\rho}{\rho_{\perp}} \left(1 - \frac{M_s^2 H'^2}{4K_u^2} \right) \right] \quad (5.80)$$

or $\rho(H) = \rho(0) - \Delta\rho(M_s^2 H'^2 / 4K_u^2)$, provided $M_s H' / 2K_u < 1$. Note the H'^2 variation, as predicted by Kohler's rule. This result applies for anhysteretic magnetization rotation. Magnetization processes that only involve motion of domain walls in a direction perpendicular to the easy axis have no effect on the magnetoresistance, whatever the angle between the current and applied field.

The reason why AMR is usually positive ($\rho_{\parallel} > \rho_{\perp}$) is related to the spin-orbit interaction, which tends to mix the $\uparrow$ and $\downarrow$ channels. Spin-flip scattering of the mobile $\uparrow$ electrons is permitted, and the scattering is effective when the plane of the orbit lies in the plane of the current, but not when it lies in the perpendicular direction.

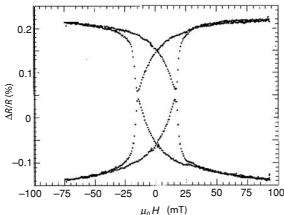
A phenomenon related to AMR, which depends on the direction of magnetization of a conducting ferromagnet, is the planar Hall effect. Amagnetic field is



Calculation of the planar Hall effect.

Figure 5.40

Field dependence of magnetoresistance for a nickel film measured in the longitudinal ($\rho_{\parallel}$) and transverse ($\rho_{\perp}$) direction (M. Viret, D. Vignoles, D. Cole *et al.*, *Phys. Rev. B* **53**, 8464 (1996)).



applied in the plane of a thin sample, transverse to the direction of the current j_x , which is in the x -direction. If the magnetization lies in-plane at an angle φ to the current direction, a transverse electric field E_y appears, just as in the normal Hall effect. The anisotropic resistivity is related to the direction of M . Resolving the electric field E into components parallel and perpendicular to M , we have

$$E_{\parallel} = \rho_{\parallel} j_x \cos \varphi; \quad E_{\perp} = \rho_{\perp} j_x \sin \varphi.$$

Now the components of the electric field parallel and perpendicular to the current are $E_x = E_{\parallel} \cos \varphi + E_{\perp} \sin \varphi$ and $E_y = E_{\parallel} \sin \varphi - E_{\perp} \cos \varphi$, hence

$$E_x = j_x (\rho_{\perp} + \Delta \rho \cos^2 \varphi) \quad \text{and} \quad E_y = j_x \Delta \rho \sin \varphi \cos \varphi.$$

The planar Hall voltage is therefore

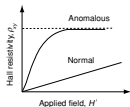
$$V_{pH} = j_x w \Delta \rho \sin \varphi \cos \varphi, \quad (5.81)$$

where w is the width of the film. The planar Hall resistivity is $\rho_{xy} = \frac{1}{2} \Delta \rho \sin 2\varphi$. The greatest change is found when the magnetization switches from $\varphi = 45^\circ$ to $\varphi = 135^\circ$.

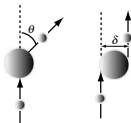
So far, we have assumed that the magnetization of the ferromagnet is uniform, and that the current flows at some fixed angle φ with M . When electrons move in a solid they are influenced by the B -field they experience. In a ferromagnet, the magnetization contributes to electron scattering, so that Kohler's rule (5.74) should be generalized to read

$$\frac{\Delta \rho}{\rho} = a(H/\rho)^2 + b(M/\rho)^2. \quad (5.82)$$

Data for nickel in Fig. 5.40 show that the resistance exhibits a butterfly hysteresis loop that reflects the hysteresis of the magnetization, although it is an even function of M , as expected.



Anomalous Hall effect.



(i) Skew scattering and (ii) side jump scattering which both contribute to the anomalous Hall effect.

Anomalous Hall effect There is a term in the Hall resistivity of a ferromagnet when the field is applied in the z -direction, perpendicular to the plane of the film, in addition to the normal Hall effect (3.53). This is the anomalous Hall effect, which varies with the magnitude of the magnetization M :

$$\varrho_{xy} = \mu_0(R_h H' + R_s M). \quad (5.83)$$

The anomalous Hall effect is yet another consequence of spin-orbit coupling. The symmetry of the radial component of the Lorentz force $\mathbf{j} \times \mathbf{B}$ which produces the normal Hall effect is the same as the symmetry of the spin-orbit interaction $\mathbf{L} \cdot \mathbf{S}$ since $\mathbf{L} = \mathbf{r} \times \mathbf{p}$, $\mathbf{p} \propto \mathbf{j}$, $\mathbf{S} \propto \mu_0 \mathbf{M}$.

In a ferromagnet the anomalous Hall effect varies as the macroscopic average magnetization. Generally there are contributions which vary as ϱ_{xx} and as ϱ_{xx}^2 , which are associated with different scattering mechanisms. Deviation of the electron trajectories due to spin-orbit interaction is known as skew scattering.

Writing $\varrho_m = \mu_0 R H'$, the Hall angle ϕ_H is defined as ϱ_m / ϱ_{xx} . Thus $\phi_H = \alpha + \beta \varrho_{xx}$; α is the skew scattering angle. The second term is often larger. It is associated with the side-jump mechanism due to impurity scattering. If $\delta \approx 0.1$ nm is the side jump, the Hall angle here is δ / λ , which is proportional to ϱ_{xx} . Here λ is the mean free path.

The anomalous Hall effect is especially large in disordered ferromagnets with strong impurity scattering, and in magnetic glasses. There is a small effect in paramagnets, which follows a Curie–Weiss law. It is most useful for measuring hysteresis in films when their Magnetization lies perpendicular to the plane.

Colossal magnetoresistance CMR is the substantial fall in resistance observed in the vicinity of the Curie temperature of double-exchange materials such as $\text{La}_{0.7}\text{Ca}_{0.3}\text{MnO}_3$ when they are subjected to a large applied magnetic field. The field aligns the partly disordered manganese moments, which promotes electron hopping (§5.2), and in turn enhances the exchange. Such ferromagnetic mixed valence oxides, where double exchange is the dominant ferromagnetic coupling mechanism, show enhanced negative magnetoresistance in the vicinity of T_C . With our definition (5.72) the limit is -100% , but an alternative, optimistic definition places $\varrho(H)$ in the denominator:

$$MR = [\varrho(H) - \varrho(0)] / \varrho(H),$$

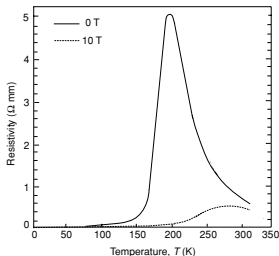
producing a bigger number for the same change. For instance, if $\varrho(H) = 0.1\varrho(0)$, a 90% MR becomes a 900% effect with the new definition. There is some justification for this, when the primary effect of magnetic field is to increase conductance, since the definition is equivalent to

$$MR = [\sigma(H) - \sigma(0)] / \sigma(0).$$

This is the order of magnitude that can be observed in materials such as $\text{La}_{0.7}\text{Sr}_{0.3}\text{MnO}_3$ in a large field, which merits the epithet ‘colossal’. Since

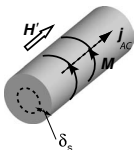
Figure 5.41

Magnetoresistance of $\text{La}_{0.7}\text{Ca}_{0.3}\text{MnO}_3$ in a field of 10 T.



electron hopping (Fig 5.10) in a mixed-valence half-metal depends on $\cos(\theta_{ij}/2)$ the resistance depends critically on the nearest-neighbour correlations $\langle \mathbf{S}_i \cdot \mathbf{S}_j \rangle \approx (1 - \theta_{ij}^2)$. Applying the field has the double effect of reducing θ_{ij} thereby increasing the ferromagnetic exchange and reducing θ_{ij} further. Some typical CMR data are shown in Fig. 5.41.

Despite the impressive magnitude of the resistance change, CMR is of limited use because of the huge fields needed to create it. If the ratio of the resistance change to the field needed to produce it is taken as the figure of merit, one is better served by an AMR of 1% in a few milliteslas, than a CMR of 90% in several teslas. The temperature dependence of resistivity near T_C in CMR materials has found some use in bolometers.



Magnetoimpedance. The AC field created by j_{AC} penetrates the wire to a depth δ_s , the skin depth. The permeability of the circumferentially magnetized wire is saturated in the applied field H' .

Magnetoimpedance The total impedance Z of a ferromagnetic component changes when it is exposed to a static magnetic field. The effect relates to the inductance L :

$$Z = R + i\omega L,$$

where $L = \Phi/I$ is the ratio of magnetic flux to current. Magnetoimpedance is conventionally defined as

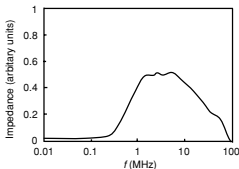
$$MI = \frac{Z(H) - Z_0}{Z_0}, \quad (5.84)$$

where Z_0 is the impedance in a field that is large enough to saturate the effect. In soft ferromagnets, the effect may be hundreds of per cent at megahertz frequencies. Hence the 'giant' epithet again.

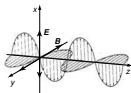
The effect is related to the skin depth δ_s (12.2), which is the depth an AC field penetrates into a metal. In a ferromagnet, δ_s varies as $(\mu_r \omega)^{-1/2}$.

Figure 5.42

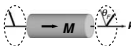
Frequency response of the impedance of an amorphous $\text{Fe}_{43}\text{Co}_{68.2}\text{Si}_{12.5}$ wire, 30 μm in diameter, in zero field, and in a field of 800 A m^{-1} applied parallel to the wire (after L. V. Panina and K. Mohri, *Appl. Phys. Lett.* **65** 1189 (1994)).



Magnetoimpedance is best explained for a soft ferromagnetic wire carrying an AC current $I = I_0 \exp(i\omega t)$. The current creates a circumferential magnetic field around the wire that penetrates to a depth δ_s . Circumferential anisotropy K_c is created if the wire has a small positive magnetostriction. The anisotropy field $H_a = 2K_c/\mu_0 M_s$ controls the response of the induced magnetization to a DC field H' applied parallel to the wire, and hence the permeability in the skin depth. For good results, H_a should be an order of magnitude greater than the field H' it is desired to measure. A typical response of a soft amorphous ferromagnetic wire is shown in Fig. 5.42.



An electromagnetic wave.



The magneto-optic Faraday effect. The plane of polarization of incident light is rotated through an angle θ_F . In a magneto-optic isolator, $\theta_F = \pi/4$, and polarizers offset by this angle are placed at each end.

5.6.5 Magneto-optics

Michael Faraday's 1845 discovery of a connection between magnetism and light was an advance comparable in importance to Hans-Christian Oersted's discovery of the connection between magnetism and electricity. He found that the plane of polarization of light rotates as it passes through a glass plate when a magnetic field is applied parallel to the direction $\mathbf{e}_K$ of propagation of the light. Light is an electromagnetic wave, where the $\mathbf{E}$ -field and the $\mathbf{H}$ -field are perpendicular to each other, and transverse to the propagation vector $\mathbf{K} = (2\pi/\Lambda)\mathbf{e}_K$; Λ is the wavelength of light. The direction of polarization is defined by $\mathbf{E}$.

An $\mathbf{H}$ -field has no influence on the electromagnetic wave; the Faraday effect is due to the interaction of light with the induced magnetization $\mathbf{M}$ of the solid medium. Faraday's original demonstration was for paramagnetic glass, but Faraday rotation is also observed when light traverses a diamagnet or a transparent, spontaneously magnetized ferromagnet or ferrimagnet. The Faraday rotation in water, for example, is $3.9 \text{ rad T}^{-1}\text{m}^{-1}$. The sense of rotation reverses when the magnetization direction is switched. Conventionally, the angle of rotation θ_F is positive if the plane of polarization of the light turns clockwise when the observer faces the source.

Table 5.12. Faraday rotation in ferromagnetic metals and ferrimagnetic insulators

	θ_F/t at 830 nm ($^\circ \mu\text{m}^{-1}$)		θ_F/t at 1.06 μm ($^\circ \text{mm}^{-1}$)
Fe	35	$\text{Y}_3\text{Fe}_5\text{O}_{12}$	28
Co	36	$\text{Tb}_3\text{Fe}_5\text{O}_{12}$	54
Ni	10		

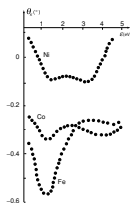
The Faraday constant is proportional to the optical path length

$$\theta_F = k_V \int \mu_0 \mathbf{M} \cdot d\mathbf{l} \quad (5.85)$$

where $k_V(\lambda)$ is the Verdet constant of the material, which depends on wavelength. Since the $\mathbf{H}$ -field has no influence, (5.85) may be equally well written as $\theta_F = k_V B l$ for a uniformly magnetized sample. The Faraday effect is nonreciprocal in the sense that the rotation (clockwise or counterclockwise) depends on whether the wavevector $\mathbf{K}$ of light is parallel or antiparallel to $\mathbf{M}$. Hence Faraday rotation is cumulative; as light is reflected back along its path in a magnetic medium, the resultant rotation is twice that for a single pass, not zero. This property is exploited in magneto-optic isolators. If the length l is chosen so that $\theta_F = \pi/4$, light entering through a polarizer passes through the isolator in one direction, but reflected light is blocked. Some values of θ_F per unit path length are given in Table 5.12. The Verdet constants are 1000 times greater for ferromagnetic metals than for transparent ferrimagnetic insulators, but the comparison is misleading because metals are opaque unless they are very thin. A better figure of merit is the product of k_V and the absorption length, which is a few degrees per tesla for both. The Faraday rotation is strongly wavelength dependent, and it is enhanced near an absorption edge.

A related effect for reflected light was discovered by John Kerr in 1877. He noticed that the plane of polarization of light reflected from the polished iron pole face of an electromagnet was rotated by less than 1° . The sense of the rotation again switched when the magnetization reversed. The Kerr rotation for 3d ferromagnets shows a maximum in the optical region at about 1.5 eV (Table 5.13).

The polar configuration is one of three where a magneto-optic Kerr effect (MOKE) can be observed. In the other two, the magnetization lies in-plane, either transverse to the plane containing the incident and reflected rays, or longitudinal within that plane (Fig. 5.43). In the longitudinal configuration, there is also a rotation of the polarization of the incident ray by less than a degree, and a slight ellipticity of the reflected beam, but in the transverse configuration there is just a difference in reflectivity for light polarized in-plane or perpendicular to the plane of incidence, which depends on the magnetization of the sample.



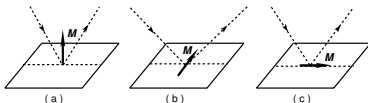
Kerr rotation as a function of photon energy for Fe, Co, Ni.

Table 5.13. Kerr rotation in ferromagnetic metals at 830 nm (1.5 eV)

	θ_K (°)		θ_K (°)
Fe	-0.53	CoPd	-0.17
FePt	-0.39	CoPt	-0.36
FeCo	-0.60	Ni	-0.09
Co	-0.36	PtMnSb	-1.3

Figure 5.43

Three configurations in which the magneto-optic Kerr or Faraday effects can be observed: (a) polar, (b) transverse, (c) longitudinal.



For the polar and longitudinal effects, there is a small, magnetization-sensitive component k_K at right angles to the reflected intensity r_K which leads to the Kerr rotation θ_K and ellipticity e_K . When $k_K \ll r_K$, θ_K and e_K are given by the expressions

$$\theta_K = \psi_K \cos \phi_K \quad \text{and} \quad e_K = \psi_K \sin \phi_K,$$

where $\psi_K = |K/r|$, and ϕ_K is the phase difference between K and r . The polar Kerr effect finds applications in measuring the hysteresis of thin films, domain imaging and magneto-optic recording.

Both Faraday and Kerr effects are related to spin-orbit coupling. They may be enhanced by incorporating heavy elements such as bismuth into the crystal lattice. Magnetic permeability is not involved because the permeability is frequency-independent and equal to μ_0 at optical frequencies, which are orders of magnitude greater than the Larmor precession or ferromagnetic resonance frequency. The speed of light in a medium, v , just depends on its electrical permittivity ϵ : $v = (\mu_0 \epsilon)^{-1/2}$, where $\epsilon = \epsilon_0 \epsilon_r$.

The permittivity of a crystal is generally a symmetric tensor defined by $D_i = \epsilon_{ij} E_j$. It is simplified in the principal axis system to one with only diagonal elements, ϵ_{xx} , ϵ_{yy} , ϵ_{zz} . When the crystal is isotropic, all three are equal. However, if the crystal is magnetized along Oz the direction of propagation of the light, the magneto-optic permittivity tensor becomes a Hermitian, skew-symmetric matrix if spin-orbit coupling is present:

$$\hat{\epsilon} = \epsilon_0 \epsilon_r \begin{pmatrix} 1 & iQ & 0 \\ -iQ & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (5.86)$$

This is an example of Onsager's principle for the symmetry of a response function $\epsilon_{ij}(M) = -\epsilon_{ji}(M) = \epsilon_{ji}(-M)$. Both the index of refraction $n = \epsilon_r^{1/2}$ and the magneto-optic parameter Q have real and imaginary parts:

$$n = n' - in'' \quad \text{and} \quad Q = Q' - iQ'',$$

which are complex constants of the material that define its magneto-optic properties.

An interpretation of the off-diagonal matrix elements is that the magnetic field $\mathbf{H}$ in the electromagnetic wave gives rise to a Lorentz force $-\mu_0 e(\mathbf{v} \times \mathbf{H})$ which mixes the x and y components of the electron's motion. In ferromagnets with quenched orbital moments, it is not possible to distinguish between clockwise and anticlockwise electron motion, and $\epsilon_{xy} = 0$. However, spin-orbit interaction restores an orbital moment, and leads to magnetic dichroism and Faraday or Kerr rotation.

An eigenvalue equation $\hat{\epsilon} \mathbf{E} = n_0^2 \mathbf{E}$ can be formed from (5.86). The eigenvalues obtained by diagonalizing the matrix are $n = \epsilon_r^{1/2}(1 \pm Q)^{1/2} \simeq \epsilon_r^{1/2}(1 \pm \frac{1}{2}Q)$ since Q is a small quantity, of order 10^{-3} or 10^{-4} . The two eigenmodes are $\begin{bmatrix} 1 \\ i \end{bmatrix}$ and $\begin{bmatrix} 1 \\ -i \end{bmatrix}$ or $E_x \pm iE_y$, which correspond to left and right circular polarization of the electric field. These modes, denoted σ^+ and σ^- , experience slightly different refractive indices, and therefore propagate at slightly different speeds $c/n = c\epsilon_r^{1/2}(1 \pm \frac{1}{2}Q)^{-1}$. This is an effect known as circular birefringence and the resulting phase difference between the modes causes the Faraday rotation.

The two fundamental modes σ^+ and σ^- correspond to the quanta of electromagnetic radiation, known as photons, which are massless particles with $\ell = 1$ and $m_\ell = \pm 1$ (but not zero). Photons carry angular momentum, which can take two values, $\pm\hbar$, when projected along the direction of propagation. The polarization of the left- and right-circularly polarized beams rotate clockwise and anticlockwise respectively, when viewing the source. When the light is absorbed by the electron system of a solid, the photons can transmit their angular momentum to the electrons, thereby changing the projection of the magnetic moment along Oz .

When $\mathbf{K} \parallel \mathbf{M}$, the orbital contribution to the magnetization involves electrons circulating in the same sense as $\mathbf{E}$ for one mode and in the opposite sense for the other. The two modes see slightly different relative permittivities $\epsilon_r^{\pm}$ and refractive indices $n^{\pm}$, where $\epsilon_r = n^2$. The absorption is also slightly different for the two modes, an effect known as circular dichroism which leads to an elliptical polarization of the rectilinearly polarized incident light.

A summary of the four main magneto-optic phenomena is provided in Table 5.14. Besides the Faraday and Kerr effects, linear in M , which depend on spin-orbit coupling, there are also second-order magneto-optic effects varying as M^2 which depend on whether the plane of polarization of the light lies parallel or perpendicular to the magnetization, which is in the xy -plane.

Table 5.14. Magneto-optic phenomena

Geometry	Physical effect	Phenomenon	Condition	M dependence
$M \parallel K$	Circular birefringence	Faraday/Kerr rotation	$n^+ \neq n^-$	$\sim M$
	Circular dichroism	Magnetic circular dichroism	$\alpha^+ \neq \alpha^-$	$\sim M$
$M \perp K$	Linear birefringence	Cotton-Mouton effect	$n^{\parallel} \neq n^{\perp}$	$\sim M^2$
	Linear dichroism	Magnetic linear dichroism	$\alpha^{\parallel} \neq \alpha^{\perp}$	$\sim M^2$

Magneto-optic effects are not restricted to the optical frequency range, but exist also in the microwave, ultraviolet and X-ray ranges. Magnetic permeability is important at microwave frequencies. Magnetic circular dichroism (MCD) is enhanced near X-ray absorption edges, especially the L -edges of transition metals. It is a useful technique for investigating magnetic materials because it is element specific, and capable of separating the spin and orbital contributions to the magnetic moment (see §10.4.2).

Finally, we mention nonlinear optical effects which appear as a result of the intense electric fields in pulsed laser beams. These effects are forbidden in centrosymmetric crystals. At surfaces, there is no centre of symmetry, and second harmonic generation is a particularly sensitive method for selectively studying the magnetization of ferromagnetic surfaces and interfaces in reflection.

FURTHER READING

- J. S. Smart, *Effective Field Theories of Magnetism*, Philadelphia: Saunders (1966). A concise summary of molecular field theory of ferromagnets, antiferromagnets and ferrimagnets.
- D. C. Matthis, *The Theory of Magnetism Made Simple*, Singapore: World Scientific (2006). The retitled new edition of an accessible theoretical text, first published in 1965.
- A. K. Zvedin and V. A. Kotov, *Modern Magnetoptics and Magneto-optical Materials*, Bristol: IOP (1997).
- A. M. Tishin and Y. I. Spichkin, *Magnetocaloric Effect and Its Applications*, Bristol: IOP (2004).
- P. Mohn, *Magnetism in the Solid State; Introduction*, Berlin: Springer (2005). Focussed on itinerant electron ferromagnetism.
- E. du Tremolet de Lachesserie, *Magnetostriction; Theory and Applications of Magnetoelasticity*, Boston: CRC Press (1993). Detailed theory of magnetostriction.
- A. del Moral, *Handbook of Magnetostriction and Magnetostrictive Materials*, 2 volumes, Zaragoza: del Moral (2008). A new and exhaustive treatment of the subject.
- R. C. O'Handley, *Modern Magnetic Materials; Principles and Applications*, New York: Wiley (1999). A thorough treatment of modern materials and applications, especially metal, alloys and thin films.

M. Knober, M. Vásquez, and L. Kraus, Giant magnetoimpedance, in *Handbook of Magnetic Materials* (K. H. J. Buschow, editor). Amsterdam: North Holland, Vol 15, p. 497–564.

EXERCISES

- 5.1 Calculate the heat capacity for a ferromagnet near T_C in Landau theory.
- 5.2 Estimate the Curie temperature of gadolinium from the RKKY model. Take $S = 7/2$ and $n = 3 \times 10^{28} \text{ m}^{-3}$.
- 5.3 Use the magnetic valence model to estimate:
 - (a) the average moment/atom in an $\text{Fe}_{40}\text{Ni}_{30}\text{B}_{20}$ alloy and in $\text{Co}_{80}\text{Cr}_{20}$;
 - (b) the critical concentration for the appearance of magnetism in amorphous $\text{Y}_{1-x}\text{Fe}_x$.
- 5.4 Diagonalize the matrix (5.39). Rank the eigenvalues in order and identify the states to which they correspond.
- 5.5 Use (5.24) and (5.57) to deduce the result that the ground state for a chain of spins has eigenvalue $-2(N-1)\mathcal{I}S^2$.
- 5.6 Show that the low-temperature heat capacity due to excitations which obey a dispersion relation $\varepsilon = Dq^2$ varies as $T^{3/2}$ for a three-dimensional solid. How is the heat capacity modified by an anisotropy gap Δ at $q = 0$?
- 5.7 By minimizing the energy of a uniaxial ferromagnet with two anisotropy constants K_1 and K_2 , show how to deduce the two constants from the magnetization curve in the hard direction (Sucksmith–Thompson plot).
- 5.8 Explain why it is impossible to produce a permanent magnet of arbitrary shape using shape anisotropy alone.
- 5.9 Show explicitly that the quadrupole moment of a rare-earth ion vanishes at high temperatures. Is it possible to observe anisotropy above the Curie point?
- 5.10 Deduce the $\kappa_j(T)/\kappa_j(0) = M^{j(j+1)/2}$ law for $j = 2, 4$ (see H. B. Callen and E. Callen; *J. Phys. Chem. Solids*, **27** 1271 (1966)).
- 5.11 Show that the spontaneous volume magnetostriction varies as M_j^2 . The associated energy is $\frac{1}{2}K\omega_v^2$, where K is the bulk modulus.
- 5.12 Discuss the effect on (a) the magnetization and (b) the resistivity of introducing 5% of Cr into a Ni host.
- 5.13 Devise a mechanical experiment to show that light carries angular momentum, making an estimate of the magnitude of the effect that might be observed.
- 5.14 The magnetization calculated from Langevin theory has nonzero slope at $T = 0 \text{ K}$. According to (2.102) this implies that the entropy is finite, which is incompatible with the third law of thermodynamics. Discuss.